Supplementary material: Full-phase complex Hamiltonian learning with conservative certification and dissipative magnetic feedback

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The proofs use the full four-coordinate learned energy and the same prescribed finite-Gauss mechanical graph as the article. The observation certificates distinguish numerical-reference fidelity from physical measurement error. The conservative-family theorem is instantiated by its completed numerical application below. The prospective independent-anchor and statistical-risk results retain their additional assumptions. The port-power and sampled physical-energy statements use separate nonconservative arguments.

S1 Local and accumulated error of the conservative midpoint step

This proof supplies explicit constants for the accuracy statement in the article. Let $J_c = \left(\begin{smallmatrix} 0 & I_2 \\ -I_2 & 0 \end{smallmatrix}\right)$ and let $f = J_c \nabla \widehat{H}$ be C^2 on an open domain. Assume a common convex neighborhood contains all states and segments used below. Let M, L, B_2 bound $f, Df, D^2 f$, respectively, in a fixed vector norm and its induced multilinear norms. Exact trajectories and the unique local midpoint branches must remain in this domain through time T. These are domain hypotheses, not conclusions from a formal order calculation.

For an exact solution $x(t)$,

$$x'' = Df(x)f(x), \qquad x''' = D^2 f(x)[f(x), f(x)] + Df(x)Df(x)f(x).$$

Consequently $\|x''\| \le LM$ and $\|x'''\| \le M_3 := B_2 M^2 + L^2 M$. For $x_0 = x(0)$ and $x_h = x(h)$, the composite defect of the exact endpoint in the midpoint equation is

$$d_h = x_h - x_0 - hf((x_0 + x_h)/2).$$

The one-panel midpoint quadrature remainder for $\int_0^h f(x(t))\,dt$ is bounded by $M_3 h^3/24$. Moreover, linear interpolation of a twice differentiable trajectory gives

$$\|x(h/2) - (x_0 + x_h)/2\| \le LMh^2/8.$$

Applying the Lipschitz constant L to this chord-midpoint discrepancy yields

$$\|d_h\| \le \left(\frac{M_3}{24} + \frac{L^2 M}{8}\right) h^3 = \left(\frac{B_2 M^2}{24} + \frac{L^2 M}{6}\right) h^3 = C_d h^3.$$

Both estimates follow by the integral Taylor remainder, so bounded continuous second derivatives of f suffice.

Let $y = \Phi_h(x_0)$ be the exact solution of the implicit midpoint equation. Subtraction from the defect equation gives

$$\|y - x_h\| \le (hL/2)\,\|y - x_h\| + \|d_h\|\,.$$

For $hL/2 \le \kappa < 1$, the one-step bound is therefore $\|y - x_h\| \le C_\ell h^3$ with $C_\ell = C_d/(1-\kappa)$.

Two initial states and their midpoint endpoints similarly satisfy

$$\|\Phi_h(x) - \Phi_h(y)\| \le \frac{1 + hL/2}{1 - hL/2} \|x - y\| \le e^{\Lambda h} \|x - y\|, \qquad \Lambda = L/(1-\kappa).$$

The last inequality follows by integrating $d \log[(1+a)/(1-a)]/da = 2/(1-a^2)$ and using $0 \le a \le \kappa$, or from $\log[(1+a)/(1-a)] \le 2a/(1-a)$. Writing $e_n = \|x_n - x(nh)\|$ gives $e_{n+1} \le e^{\Lambda h} e_n + C_\ell h^3$. For $e_0 = 0$,

$$e_n \le C_\ell h^3 \frac{e^{\Lambda n h} - 1}{e^{\Lambda h} - 1} \le C_\ell h^2 \frac{e^{\Lambda T} - 1}{\Lambda}.$$

Continuous limits cover $\Lambda = 0$. A nonzero initial error contributes the additional term $e^{\Lambda T} e_0$.

If an implementation returns $\widetilde{y}$ with algebraic residual $r = \widetilde{y} - x - hf((x + \widetilde{y})/2)$, contraction gives

$$\|\widetilde{y} - \Phi_h(x)\| \le \frac{\|r\|}{1 - hL/2}.$$

These errors enter the same accumulated-error recursion. Per-step residuals of order h^3 preserve the stated global second-order rate under the same domain bounds. They do not imply that the computed map satisfies the exact symplectic Jacobian identity.

S2 Strict finite-output inverses in the observed canonical chart

The following certificate replaces numerical rank by a strict inverse for the selected finite-output designs. It uses the prescribed finite Gauss-eight metric, with all its mechanical buffers shared exactly by the full-phase GC. At each selected moving node, interpret the stored binary64 $q_{\mathrm{obs}}, v_{\mathrm{obs}}$ as exact input numbers and define

$$p_{\mathrm{obs}} = G(q_{\mathrm{obs}}) v_{\mathrm{obs}}, \qquad x_{\mathrm{obs}} = (q_{\mathrm{obs}}, p_{\mathrm{obs}}).$$

The metric and this momentum are enclosed by direct interval evaluation of the finite graph, rather than by a midpoint replacement. These points need not be the exact states of the native initial-value problem. For the specified reference with known kinetic energy and configuration-only unknown correction, they nevertheless satisfy

$$H_{*,p}(x_{\mathrm{obs}}) = v_{\mathrm{obs}}, \qquad (H_\theta - H_*)_p(x_{\mathrm{obs}}) = N_{\theta,p}(x_{\mathrm{obs}}).$$

Thus the momentum-error label at these exact constructed points is structurally zero. No additional equality between p_{obs} and true native momentum is needed to test the function $N_{\theta,p}$ there. Noise or trajectory-error budgets are required if a subsequent claim transfers the result to true physical states or dynamics; this certificate does not declare those budgets zero.

Let $\chi_\alpha(x) = \prod_{j=1}^4 (x_j/s_j)^{\alpha_j}$, where the s_j are the exact four GC scale buffers. Physical coordinate derivatives retain the factor $1/s_j$. For total even degrees two and four, use the selected eight rest configuration-gradient rows on the configuration-only columns and the selected 37 moving momentum-gradient rows on the other columns. For degrees two, four and six the corresponding dimensions are 15 and 114. A column permutation places each full selected design in the form

$$A_{45} = \mathrm{diag}(A_{q,8}, A_{p,37}), \qquad A_{129} = \mathrm{diag}(A_{q,15}, A_{p,114}).$$

The cross blocks vanish by polynomial differentiation: rest configuration derivatives leave all momentum factors intact, and momentum derivatives annihilate configuration-only monomials. There is no additional row weighting or column equilibration in these strict matrices.

Lemma S1. Dyadic approximate inverse certificate. *Suppose an interval matrix encloses the exact square design A, and let R be a specified exact dyadic matrix. Define $E = I - RA$. If*

$$\|E\|_\infty \le \eta_\infty < 1,$$

then A is nonsingular and

$$\|A^{-1}\|_\infty \le \frac{\|R\|_\infty}{1-\eta_\infty}.$$

If also $\|E\|_1 \le \eta_1 < 1$ and $\eta_2 \ge \sqrt{\eta_1\eta_\infty}$ satisfies $\eta_2 < 1$, then

$$\|A^{-1}\|_1 \le \frac{\|R\|_1}{1-\eta_1},$$
$$\|A^{-1}\|_2 \le \sqrt{\frac{\|R\|_1\|R\|_\infty}{(1-\eta_1)(1-\eta_\infty)}},$$
$$\|A^{-1}\|_F \le \frac{\|R\|_F}{1-\eta_2}.$$

Each square-root bound may be enlarged by directed rational rounding.

Proof. The Neumann series gives $(I-E)^{-1}$ and its induced-norm bounds. Since $RA = I - E$ is nonsingular and the matrices are square, A is nonsingular and $A^{-1} = (I-E)^{-1}R$. Use $\|E\|_2 \le \sqrt{\|E\|_1\|E\|_\infty}$, $\|BC\|_F \le \|B\|_2\|C\|_F$ and $\|A^{-1}\|_2 \le \sqrt{\|A^{-1}\|_1\|A^{-1}\|_\infty}$. The alternative bounds $\|A^{-1}\|_F \le \sqrt{n}\|A^{-1}\|_\infty$ and $\|A^{-1}\|_F \le \sqrt{n}\|A^{-1}\|_1$ may further tighten the result.

The computation uses 512-bit outward interval arithmetic for the direct metric, monomials and complete matrix product. An ordinary midpoint inverse merely proposes R; each entry is then rounded onto the exact 2^{-96} grid. The proof uses only the full interval enclosure of $I - RA$, not the accuracy of that floating inversion. All selected input words, row identities, basis exponents, scales, matrix enclosures and dyadic inverse entries are retained.

Block	η_∞ upper	$\|A^{-1}\|_\infty$ upper	$\|A^{-1}\|_F$ upper
$A_{q,8}$	$5.616\,10^{-15}$	0.207182	0.182883
$A_{p,37}$	$1.737\,10^{-13}$	1.013895	0.477587
$A_{q,15}$	$5.446\,10^{-14}$	1.781184	1.044917
$A_{p,114}$	$2.594\,10^{-9}$	514.632787	113.561027

Consequently the selected designs have exact ranks 45 and 129. For a block diagonal inverse, the infinity and spectral norms are the maximum of the block norms, and squared Frobenius norms add. Row or column permutations do not change these norms. The resulting combined infinity bounds are less than 1.013895 and 514.632787 respectively.

For an exact coefficient identity $z = Aa + t - \xi$, the inverse now supplies the nontrivial finite-output step

$$a = A^{-1}z + A^{-1}(\xi - t).$$

Actual signed residuals, pointwise tail bounds and label/evaluation allowances must still be inserted before concluding a learning-error bound. The zero reference label for N_p at the constructed points does not assert that the learned N_p, its tail, or its evaluation error is zero. The q-gradient rows retain their own nonzero reference label-error budget. Neural analytic remainders outside this finite representative, the appropriate real and complex domains, and any KAM acceptance remain separate certificates. This inverse does not turn dependent moving nodes into iid samples, certify native solver accuracy, or establish identification of an unknown physical plant.

S3 Finite-output observation acceptance for a full-phase Hamiltonian

Let the same known kinetic graph be used in

$$H_*(q,p) = T_G(q,p) + V_4(q) + U(q), \qquad H_\theta(q,p) = T_G(q,p) + V_4(q) + N_\theta(q,p), \quad T_G = \tfrac{1}{2}p^{\mathsf{T}}G(q)^{-1}p.$$

Fix the physical canonical coordinates $x = (q_1, q_2, p_1, p_2)$ and their positive stored scales s_j. For $D = 4$ or 6, use all monomials

$$\chi_\beta(x) = \prod_{j=1}^{4}(x_j/s_j)^{\beta_j}, \qquad |\beta| = 2, 4, \ldots, D.$$

There are $n = 45$ or 129 columns. This is a finite representative of an output, not a restriction of the neural-network hypothesis class. For each admitted output choose an explicitly identified representative and define its actual remainder by

$$P_\theta = \chi^{\mathsf{T}} a_\theta, \qquad \rho_\theta = N_\theta - P_\theta, \qquad E_\theta = N_\theta - U = \chi^{\mathsf{T}} a_\theta + \rho_\theta - U. \tag{S1}$$

No reference coefficient is used as a network training label or as an assumed value of a_θ.

Exact observation anchors and row units. At rest anchors $(Q_i, 0)$, let sealed potential-gradient labels satisfy

$$Y_i = \nabla(V_4 + U)(Q_i) + \xi_i.$$

The label-fidelity allowance must include whatever initial-rest, finite-difference, actual-clock, observation and numerical errors occur in its construction. It is not replaced by a solver tolerance. For moving records, define the exact observed-state chart

$$z_l = (\bar{q}_l, \bar{p}_l), \qquad \bar{p}_l = G(\bar{q}_l)\bar{v}_l, \tag{S2}$$

where $\bar{q}_l, \bar{v}_l$ are the exact represented observation numbers and G is the prescribed exact graph. Since $H_{*,p}(z_l) = \bar{v}_l$, the reference momentum-gradient correction is zero at these anchors, irrespective of whether they are true IVP states. This is structural information from the shared kinetic model, not an observation of native momentum or force.

Let $\mathcal{L}$ consist of chosen q-derivative rows at rest and chosen p-derivative rows at moving anchors. Form

$$r_{q,i} = N_{\theta,q}(Q_i, 0) + V_{4,q}(Q_i) - Y_i, \qquad r_{p,l} = N_{\theta,p}(z_l). \tag{S3}$$

Only the selected components of these vector expressions are retained. The V_4 derivative is that of the actual common scalar mechanical potential; a rounded surrogate stiffness matrix needs a separate error allowance. No reference ∇U is used to form the signed residual. With ξ containing the selected rest errors and zero moving entries,

$$r = \mathcal{L}E_\theta - \xi. \tag{S4}$$

Interval evaluation may enclose this exact r instead of returning a single number. Its numerical width is distinct from physical label error.

Physical derivative rows contain $1/s_j$. If different row units or weights are desired, declare a positive diagonal matrix W and set $\widetilde{\mathcal{L}} = W\mathcal{L}$, $\widetilde{A} = WA$ and $\widetilde{r} = Wr$, where $A_{ij} = \mathcal{L}_i\chi_j$. The same W multiplies noise and point remainders. A normalized-coordinate derivative uses row weight s_j. Counts of dependent curve nodes are not weights justified by iid sampling. The present square inverse construction uses physical rows, $W = I$, and no extra column scaling.

After a recorded column permutation, the exact design is

$$A = A_q \oplus A_p, \qquad (n_q, n_p) = (8, 37) \text{ or } (15, 114).$$

Rest configuration derivatives leave every momentum factor zero, and momentum derivatives annihilate all configuration-only monomials. This structural statement uses the literal rest anchor $p = 0$; an uncertain nonzero rest momentum cannot silently be inserted while retaining the same zero blocks. Column permutations must also be applied to a_θ, every output derivative row and every inverse artifact.

Observation-chart and input errors. A numerical $G\bar{v}$ is only an approximation to (S2). Enclose the exact chart jointly in the design A and the model residual r. If instead a certificate is intended at true physical anchors, a common state-error contract is needed. For example, if $|q^* - \bar{q}| \le d_q$ and $|v^* - \bar{v}| \le d_v$, then on a containing real segment

$$|p_i^* - \bar{p}_i| \le \sum_j G_{ij}^{\mathrm{abs}} d_{v,j} + \sum_{j,k} G_{ij,k}^{(1),\mathrm{abs}} |\bar{v}_j| d_{q,k}.$$

Add $\sum_j \varepsilon_{G,ij}|\bar{v}_j|$ if the nominal chart uses an approximate metric rather than the exact graph. Derivative observations shifted between two anchors obey the componentwise allowance

$$|\partial_j E_\theta(x) - \partial_j E_\theta(\bar{x})| \le \sum_k \sup_{[x,\bar{x}]} |\partial_{jk} E_\theta|\, d_{x,k}.$$

For this transport, sound candidate-specific bounds on $|\partial_{jk} N_\theta| + |\partial_{jk} U|$ suffice; the desired small learning error is not assumed to prove itself. All segments and evaluation boxes must lie in their certified domains. State uncertainty may instead be enclosed directly when evaluating A and r. These are alternative bookkeeping conventions: a state mismatch must be covered, but need not be charged twice. In particular, the exact observed-chart inverse does not alone certify true physical-state error.

Lemma S2. Strict design inverse and signed finite-output identity. *Suppose an interval matrix encloses the actual $\widetilde{A}$ and a fixed exact dyadic matrix R satisfies*

$$\sup_{\widetilde{A}} \|I - R\widetilde{A}\|_\infty \le \eta < 1.$$

For each actual matrix define $J = (R\widetilde{A})^{-1}R$ and $\kappa = \eta/(1-\eta)$. Then $J\widetilde{A} = I$ and $\|(R\widetilde{A})^{-1} - I\|_\infty \le \kappa$. Let

$$t_N = \widetilde{\mathcal{L}}\rho_\theta, \qquad t_U = \widetilde{\mathcal{L}}U, \qquad F_\alpha(x) = \partial^\alpha \chi(x)^{\mathsf{T}}.$$

For every multi-index α for which the terms exist,

$$\begin{aligned}\partial^\alpha E_\theta(x) = {} & F_\alpha(x)J\widetilde{r} + F_\alpha(x)JW\xi \\ & + \big[\partial^\alpha \rho_\theta(x) - F_\alpha(x)Jt_N\big] + \big[F_\alpha(x)Jt_U - \partial^\alpha U(x)\big].\end{aligned} \tag{S5}$$

Proof. The Neumann series gives the inverse and the stated bound. Applying the observation operator to (S1), then using (S4), gives $\widetilde{A}a_\theta = \widetilde{r} + W\xi - t_N + t_U$. Apply J, differentiate the decomposition and substitute.

This identity keeps every reference and network remainder term. It is often preferable to enclose their sum directly as

$$C_{\alpha,\theta}(x) = \partial^\alpha(\rho_\theta - U)(x) - F_\alpha(x)J\widetilde{\mathcal{L}}(\rho_\theta - U),$$

preserving cancellation. If $U = P_U + \rho_U$ with P_U in the finite span, then $J\widetilde{\mathcal{L}}P_U$ is exactly its coefficient vector. Thus P_U cancels from the final reference remainder. The degree-six term of U belongs to the 129-column representative, but not to the 45-column representative. Replacing J by R does not preserve this exact cancellation unless the $J - R$ correction is retained.

Uniform derivative gains and computable bounds. Let Ω be the target complex rectangle, with $|x_j| \le r_j$; its modulus bounds must include both real and imaginary half-widths. A valid column envelope is

$$g_{\alpha,\beta} = \prod_{j=1}^{4} \frac{(\beta_j)_{\alpha_j}\, r_j^{\beta_j - \alpha_j}}{s_j^{\beta_j}},$$

with the term zero if some $\beta_j < \alpha_j$. Put $G_\alpha = \sum_\beta g_{\alpha,\beta}$ and, for $v \ge 0$, define

$$\mathcal{G}_\alpha(v) = \sum_\beta g_{\alpha,\beta} v_\beta + G_\alpha \kappa \|v\|_\infty.$$

Then an enclosure of the signed vector $R\widetilde{r}$ gives

$$\sup_\Omega |F_\alpha J\widetilde{r}| \le \mathcal{G}_\alpha(|R\widetilde{r}|).$$

The matrix-vector sum is enclosed *before* absolute values are taken. This is the signed principal term. If $|W\xi| \le \zeta$ on a common fidelity event, its contribution is bounded by $\mathcal{G}_\alpha(|R|\zeta)$. A separately certified operator $\beta_{\alpha,2} \ge \sup_\Omega \|F_\alpha J\|_2$ permits

the alternative bound $\beta_{\alpha,2}\|W\xi\|_2$; the minimum of two sound bounds may be used. Pointwise matrix-entry enclosures and input errors must refer to this same row order and scaling.

For a sound bound $c_{\alpha,\theta} \geq \sup_\Omega |C_{\alpha,\theta}|$, one therefore obtains the explicit acceptance inequality

$$\|\partial^\alpha E_\theta\|_\Omega \leq b_{\alpha,\theta} := \mathcal{G}_\alpha(|R\widetilde{r}|) + \mathcal{G}_\alpha(|R|\zeta) + c_{\alpha,\theta}. \tag{S6}$$

A conservative alternative bounds each bracket in (S5) separately, using its target-domain derivative bound and its transported observation-point remainder bound. Dropping either point tail, final tail or $J - R$ term is not valid. The observation points extend to a larger real domain than Ω; their point remainders need their own domain certificate. The selected network's 69 size bounds do not by themselves supply a small polynomial remainder, and are not class-uniform bounds.

For all orders $1 \leq |\alpha| \leq 3$, use the resulting component bounds to form full four-coordinate canonical increments

$$\delta_0 = \max_i b_{\mathbf{e}_i}, \qquad \delta_1 = \max_i \sum_j b_{\mathbf{e}_i+\mathbf{e}_j},$$

$$\delta_{t1} = \sum_{i,j} b_{\mathbf{e}_i+\mathbf{e}_j}, \qquad \delta_2 = \max_i \sum_{j,k} b_{\mathbf{e}_i+\mathbf{e}_j+\mathbf{e}_k}.$$

The sums retain ordered multiplicities. These are the learning-error inputs to the separately verified full-phase parameter-family KAM bridge. They do not replace the actual family invariance defect. Joint complex domains, explicit varying embeddings, complete angle/parameter tails, nonresonance, geometry, mean torsion and all scalar preservation gates must still be certified for the same output and the family $H_\lambda = H_\theta + \lambda(H_* - H_\theta)$.

Theorem S3. Simultaneous accepted outputs and no false acceptance. *Assume a common observation-fidelity event $\mathcal{E}$ of probability at least $1 - \delta$ supplies the stated label and input-state allowances for all eligible rows. The design and its strict inverse are independent of the neural candidate, although they may depend on the observed input geometry. Every candidate must have its own sound model-evaluation, representative-remainder and domain certificates. On $\mathcal{E}$,* (S6) *holds simultaneously for every candidate passing these tests, including outputs selected using the same data. If acceptance also requires the complete parameter-family KAM hypotheses with these error budgets, every accepted output has the stated physical-conservative-endpoint KAM conclusion. Consequently, for any measurable certificate-based procedure that returns an output or rejects,*

$$\mathbb{P}\{\textit{the procedure accepts an output with a false certified derivative or KAM conclusion}\} \leq \delta.$$

Proof. On the common event, the preceding deterministic identities and bounds hold for each soundly certified output, irrespective of its selection rule. The family bridge is then an ordinary deterministic implication. A false accepted conclusion is therefore contained in $\mathcal{E}^c$. No union over candidate parameters is required.

This is a pathwise accepted-output theorem, not a uniform population-risk bound for all raw networks, a guarantee of optimization success or a lower bound on acceptance probability. A common sound noise event must actually be established; a model-specific post-fit remainder cannot replace it. For fixed numerical-reference fidelity bounds the statement can be entirely deterministic, rather than assigning an invented failure probability. Neither fourteen dependent training curves nor the row counts 80 and 22428 become iid phase samples.

For a future independent configuration draw with the previously fixed selectors, the separate finite-q excitation theorem gives $\delta_q = 18/2^{48}$ for eight configurations or $53/2^{48}$ for fifteen. A joint fidelity-and-q-rank event has probability at least $1 - \delta - \delta_q$. This probability does not guarantee a well-conditioned mixed block, small tails or a KAM pass. If strict inverse certification is required before acceptance, rank failure causes rejection and adds no false-acceptance probability. The archived forty-point QR choice has its own exact rank witness; the future-draw probability is not a retrospective claim about that choice. Finite-grid laws still need a separate quantitative continuous-law coupling for a statistical theorem requiring positive phase density.

S4 Prospective excitation and sample counts for accepted full-phase outputs

A quantitative learning–KAM connection requires information in every learned canonical direction. The present construction supplements the deterministic observed-chart inverse with a quantitative prospective sampling result. It does not assign an independent sampling law to the archived forced trajectories. It also does not assert that a training algorithm finds an acceptable output.

Fix positive physical scales $s = (s_1, s_2, s_3, s_4)$ before collecting observations and let $u_j = x_j/s_j$. For $D = 4$ or 6, split all nonconstant even-total-degree monomials of degree at most D into configuration-only columns χ_q and columns containing at least one momentum factor χ_p. Their dimensions are $(8, 37)$ or $(15, 114)$. These are representatives used in a certificate; the neural Hamiltonian itself retains its full four-coordinate architecture.

At rest anchors $x = (s_1u_1, s_2u_2, 0, 0)$, observe both configuration derivatives with normalized rows $s_j\partial_{x_j}$, $j = 1, 2$. The mixed columns vanish in these rows. At moving anchors $x = s \odot u$, use both normalized momentum rows $s_j\partial_{x_j}$, $j = 3, 4$; configuration-only columns vanish in these rows. For the prescribed shared kinetic model, $v = G(q)^{-1}p$ gives the structural reference label $U_p = 0$. This is a sampling protocol for canonical anchors, not a claim that arbitrary snapshots of a physical trajectory have this law. Enforcing such anchors experimentally, or transporting from another sampling law, is an additional data requirement.

Let $b = q, p$, $d_b = \dim \chi_b$, and let $B_b(u)$ be the two-row normalized derivative matrix on the corresponding cube $\mathcal{C}_q = [-1, 1]^2$ or $\mathcal{C}_p = [-1, 1]^4$. Define

$$M_b = \mathbb{E}_{\mathrm{unif}(\mathcal{C}_b)} B_b^{\mathsf{T}} B_b, \qquad K_b(u) = \mathrm{tr}(M_b^{-1} B_b(u)^{\mathsf{T}} B_b(u)).$$

These fixed population matrices are positive definite. Indeed, vanishing rest gradient forces a configuration polynomial to be constant, while vanishing momentum gradient forces a mixed polynomial to be configuration-only. Both possibilities are excluded by the specified column sets. The rational arithmetic certificate independently checks positive elimination pivots, the exact inverse and $\mathbb{E}K_b = d_b$. All moments use $\mathbb{E}u^a = 0$ for odd a and $1/(a + 1)$ for even a.

Lemma S4. Conditioning under independent anchors. *Draw n_b independent anchors from a fixed density π_b relative to the uniform probability measure, where $\pi_b > 0$ almost everywhere. Use the row multiplier $\pi_b(u)^{-1/2}$ and suppose*

$$K_b(u)/\pi_b(u) \le L_b \quad \textit{almost everywhere.}$$

For $0 < \varepsilon_b < 1$, with failure probability at most

$$d_b \exp[-\varepsilon_b^2 n_b/(2L_b)],$$

the weighted design A_b satisfies

$$n_b^{-1} A_b^{\mathsf{T}} A_b \succeq (1 - \varepsilon_b) M_b. \tag{S7}$$

The event is simultaneous over all coefficient vectors and hence over all neural outputs having sound representatives and remainder certificates.

Proof. Apply the lower matrix Chernoff bound to the independent positive semidefinite summands

$$X_i = M_b^{-1/2} B_b(u_i)^{\mathsf{T}} B_b(u_i) M_b^{-1/2}/\pi_b(u_i).$$

Each has expectation I, and its largest eigenvalue is at most its trace, bounded by L_b. Corollary 5.2 and Remark 5.3 of Tropp, *User-Friendly Tail Bounds for Sums of Random Matrices*, https://tropp.caltech.edu/papers/Tro11-User-Friendly-preprint.pdf, give the stated bound. No independence between the two rows of one anchor is assumed; their matrix product is a single summand.

The code supplies a uniform-cube bound $K_b \le \overline{K}_b$ by an exact tensor Bernstein expansion of the leverage polynomial in $t_j = u_j^2 \in [0,1]$. Uniform sampling permits $L_b = \overline{K}_b$. A more economical full-support protocol fixes

$$\pi_b(u) = \tfrac{1}{2}[1 + K_b(u)/d_b], \qquad \tfrac{1}{2} \le \pi_b(u) \le \tfrac{1}{2}[1 + \overline{K}_b/d_b]. \tag{S8}$$

It is normalized because $\mathbb{E}K_b = d_b$, and $K_b/\pi_b = 2d_bK_b/(d_b + K_b) \le 2d_b$. Thus $L_b = 2d_b$ is valid. Rejection sampling from the uniform cube uses the explicit upper density envelope, with no physical solve needed for rejected proposals. The protocol and all scales must be fixed before the draw. Actual draws and accepted observations have not been produced here.

For $\varepsilon_q = \varepsilon_p = 1/2$, allocating failure probability $1/40$ to each block gives the explicit sufficient count

$$n_b = \lceil 8L_b \log(40d_b) \rceil .$$

The arithmetic certificate lists uniform and mixture counts separately. They bound design conditioning, not the number of observations required for successful optimization or a necessary sample lower bound.

Theorem S5. Simultaneous derivative transport for accepted outputs. *For each output write the actual error as*

$$E_\theta = N_\theta - U = \chi_q^{\mathsf{T}} c_{q,\theta} + \chi_p^{\mathsf{T}} c_{p,\theta} + w_\theta.$$

The reference representative, if used in this identity, is for verification only and is not supplied as a training label. At each sample use the weighted normalized derivative rows above. Suppose the corresponding observed residual, common label-fidelity error, and actual remainder are r_b, ξ_b and $\mathcal{L}_b w_\theta$, with

$$r_b = A_b c_{b,\theta} + \mathcal{L}_b w_\theta - \xi_b.$$

Let the sound allowances be

$$R_b = \|r_b\|_2/\sqrt{n_b}, \quad Z_b \ge \|\xi_b\|_2/\sqrt{n_b}, \quad T_{b,\theta} \ge \|\mathcal{L}_b w_\theta\|_2/\sqrt{n_b}.$$

On the two conditioning events of Lemma S4,

$$\|M_b^{1/2} c_{b,\theta}\|_2 \le (R_b + Z_b + T_{b,\theta})/\sqrt{1-\varepsilon_b}.$$

For a target real or complex domain Ω, let $F_{\alpha,b}(x) = \partial_x^\alpha \chi_b(x)^{\mathsf{T}}$ and certify

$$C_{\alpha,b} \ge \sup_{x\in\Omega} \big(F_{\alpha,b} M_b^{-1} F_{\alpha,b}^*\big)^{1/2}, \qquad W_{\alpha,\theta} \ge \sup_{\Omega} |\partial_x^\alpha w_\theta|.$$

Here $*$ conjugates the target-domain row, and the derivatives retain the physical factors $1/s_j$. Then, simultaneously for every soundly certified output, including one selected from the same observations,

$$\|\partial_x^\alpha E_\theta\|_\Omega \le \sum_{b=q,p} C_{\alpha,b} \frac{R_b + Z_b + T_{b,\theta}}{\sqrt{1-\varepsilon_b}} + W_{\alpha,\theta}. \tag{S9}$$

The stated probability is at least $19/20$, reduced by any separately allocated failure probability of the common fidelity event.

Proof. The residual identity, the triangle inequality and (S7) imply the coefficient bound. Differentiate the decomposition and apply the ordinary complex Cauchy–Schwarz inequality to $F_{\alpha,b} M_b^{-1/2} M_b^{1/2} c_{b,\theta}$. Both matrix events hold for all coefficient vectors, so adaptively choosing an output requires no union bound over network parameters. Candidate-dependent tails must actually be verified; they are not discarded on account of this simultaneity.

For $1 \le |\alpha| \le 3$, these derivative budgets feed the same full-phase family bridge as the deterministic observed-chart bounds. A KAM conclusion additionally requires actual family embeddings, complete angle and parameter tails, nonresonance, varying geometry and mean torsion, and all scalar preservation gates. This theorem provides a quantitative

prospective excitation and interpolation component; it does not retrospectively establish iid population risk for the existing data, remove physical-label uncertainty, or replace those family certificates. The Rademacher bound for the full neural class in Chen et al. (2022) and a guarantee of successful training are distinct claims and are not inferred from this finite-output result.

S4.1 A certified invariant torus of the full-phase learned Hamiltonian

Let $\widehat{H}_{\mathrm{GC}}$ denote the selected two-hidden-layer complex Hamiltonian, with its stored binary64 buffers interpreted as exact real constants and its smooth parameter constraints evaluated in exact real arithmetic. In physical canonical coordinates $x = (q_1, q_2, p_1, p_2)$,

$$\widehat{H}_{\mathrm{GC}}(q,p) = \tfrac{1}{2} p^{\mathsf{T}} G_8(q)^{-1} p + V_4(q) + N_{\mathrm{GC}}(q,p).$$

Here G_8 is the prescribed finite Gauss–Legendre mechanical graph, V_4 is the fourth-order potential, and N_{GC} is the trained full-phase correction. The correction includes its prescribed real readout, total evenization and constant energy gauge. Its real-analytic extension treats the four canonical coordinates independently; it retains both configuration and momentum derivatives.

Write $\varphi = (1+\sqrt{5})/2$ and fix the period-one rotation vector

$$\nu = \frac{\Omega_1}{2\pi}(1,\alpha), \qquad \Omega_1 = \frac{499974648628101}{500000000000000}, \qquad \alpha = \frac{2524851\varphi + 622102}{410497\varphi + 101143}. \tag{S10}$$

All strip and matrix norms below follow the entrywise analytic supremum convention: $\|A\|_\rho = \max_i \sum_j \sup_{\mathbb{T}^2_\rho} |A_{ij}|$.

Theorem S6. Fixed full-phase learned Hamiltonian. *For the frozen selected GC checkpoint and the finite Gauss mechanical graph specified by the accompanying reproducibility certificate, there exists an analytic invariant torus with rotation vector* (S10). *More precisely, the saved 8064-mode rational Fourier polynomial K_{128} admits a real-analytic invariant embedding with a holomorphic extension K_∞ to $\mathbb{T}^2_{1/400}$ satisfying*

$$J_4 \nabla \widehat{H}_{\mathrm{GC}}(K_\infty) = DK_\infty \nu, \qquad \|K_\infty - K_{128}\|_{1/400} < 9.018 \times 10^{-24}.$$

The restriction of K_∞ to the real torus is a real embedding. The image remains in the certified complex canonical domain

$$|\operatorname{Re} x_j| < 3/50, \qquad |\operatorname{Im} x_j| < 1/100, \qquad j = 1,\dots,4.$$

Proof. The exact-raw network and mechanical certificates establish a real-type holomorphic Hamiltonian on the stated domain, together with the required first, second and third Hamiltonian derivative bounds. The continued fraction in (S10) has bounded positive partial quotients and a golden tail. The all-integer arithmetic certificate therefore supplies

$$|k\cdot\nu| \geq \gamma/|k|_1 \quad (k \in \mathbb{Z}^2 \setminus \{0\}), \qquad \gamma = \frac{4828522443384859}{5764607523034234 88}.$$

The corresponding uniform cohomological bound is valid with exponent $\tau = 1$ and $c_R = 1/7$ at every analytic strip loss.

The coefficients of K_{128} satisfy exact conjugate reality and total oddness. Direct Fourier majorants establish its complex-domain inclusion, Gram inverse and real embedding. Put

$$L = DK_{128}, \quad B = (L^{\mathsf{T}}L)^{-1}, \quad N = J_4 L B, \quad T = N^{\mathsf{T}} J_4 \big(J_4 D^2 \widehat{H}_{\mathrm{GC}}(K_{128}) N - D_\nu N\big).$$

The transpose is the analytic transpose. All four canonical coordinates and the complete 4×4 Hamiltonian Hessian enter these expressions. Whole-strip bounds at $\rho_T = 7/200$, a 96×96 real quadrature, and the infinite alias estimate

$$|\langle T\rangle_{ij} - (T_{96})_{ij}| \leq \frac{4M_T e^{-2\pi\rho_T 96}}{(1-e^{-2\pi\rho_T 96})^2} < 1.686\times 10^{-6}$$

give an invertible continuous mean with $\|\langle T\rangle^{-1}\|_\infty < 19026.13$. Every saved Fourier mode is retained with its original derivative multiplier when folded onto the quadrature grid.

For $E = J_4\nabla\widehat{H}_{\mathrm{GC}}(K_{128}) - DK_{128}\nu$, a separate outer-strip proof at $\widehat{\rho} = 1/10$ bounds the composed field by $M_X < 55.544$. It uses full first-affine Fourier bounds, a complete second-preactivation interval cover on the four imaginary corners, the separate-variable maximum principle and a certified Cauchy tube. An outward-arithmetic 448×448 discrete Fourier transform then encloses the interpolation polynomial on $\rho = 1/40$. Its weighted componentwise coefficient sums and the full analytic alias-and-tail bound yield

$$\|E\|_{1/40} < 1.196 \times 10^{-42}.$$

The finite polynomial $DK_{128}\nu$ is represented exactly within the signed interpolation cube. All numerical sample enclosures, including the independent real-metric derivative remainders, enter the transform. Odd-parity Nyquist modes are retained.

Apply the canonical $n = d = 2$ case of Haro and Luque (2019, Theorem 3.1), with the audited arXiv version 1 conventions. Set $\rho_\infty = 1/400$, $a_1 = 2$ and choose each auxiliary σ as $11/10$ times its corresponding outward certified bound. These strict positive margins satisfy the theorem's hypotheses; all constants and all eight preservation and contraction inequalities are recomputed with these parameters. Their maximum quotient is less than 6.942×10^{-4}, with a sufficient residual threshold greater than 1.722×10^{-39}. The resulting displacement is less than 9.018×10^{-24}, lies within the certified domain margin, and preserves the real embedding by the normalized phase estimate. The sign adjustment between the saved normal frame and the theorem's canonical convention changes the sign of T and leaves its mean inverse norm unchanged. The theorem gives the asserted torus.

This statement concerns the specified learned conservative Hamiltonian. Its data-to-model population guarantees, dissipative trajectories and magnetic feedback require their respective learning and power-transport estimates. The finite-Gauss reference and the declared coordinate normalization remain part of the model specification.

S5 Uniform geometry and scalar gates for the actual conservative family

Keep the exact four canonical coordinates, frozen GC checkpoint, finite Gauss mechanical graph and prescribed period-one frequency of the learned torus certificate. Set

$$H_\lambda = H_{\mathrm{GC}} + \lambda(U - N), \quad P_\lambda = K_0 + V_\lambda, \quad V_\lambda = \sum_{j=1}^{m} \lambda^j U_j, \qquad 0 \le \lambda \le 1.$$

Every response is its actually saved rational Fourier polynomial. A change to an earlier U_j requires reconstruction of the subsequent response chain, its residual and all bounds in this section.

Matrix norms are the entrywise analytic supremum induced row-sum norm; transpose norms are bounded separately. At a selected strip let v, d, d^t, w, w^t bound respectively

$$\|V_\lambda\|, \quad \|DV_\lambda\|, \|(DV_\lambda)^{\mathsf{T}}\|, \quad \|D_\nu DV_\lambda\|, \quad \|(D_\nu DV_\lambda)^{\mathsf{T}}\|.$$

These follow uniformly for real $\lambda \in [0, 1]$ by summing the actual weighted Fourier majorants of all U_j. The period-one directional multiplier is $2\pi|k \cdot \nu|$; with the specified algebraic frequency it equals $\Omega_1|k_1 + \alpha k_2|$. No sampled maximum substitutes for these complete analytic-strip norms.

Domain and Hamiltonian derivatives. At $\rho = 1/40$, subtract each componentwise response majorant from the certified four-coordinate image margins of K_0 in $|\operatorname{Re} x_i| < 3/50$, $|\operatorname{Im} x_i| < 1/100$. A positive remaining margin supplies an open theorem tube; the calculator uses half the remaining radius as a strict reserve. Let $(\delta_0, \delta_1, \delta_{t1}, \delta_2)$ be the accepted full-phase data-to-output bounds for $Y = J\nabla(U - N)$, its derivative, its transposed derivative and its ordered bilinear second derivative on this same rectangle. Then

$$c_0 = \widehat{c}_0 + \delta_0, \quad c_1 = \widehat{c}_1 + \delta_1, \quad c_{t1} = \widehat{c}_{t1} + \delta_{t1}, \quad c_2 = \widehat{c}_2 + \delta_2$$

are valid uniform real-parameter field inputs. These statements do not extend an inner rectangle estimate to the larger residual angle strip. The separate angle–parameter product-domain proof is still needed for the residual Cauchy tail.

Lemma S7. Finite norm transport. *Put* $L_0 = DK_0$, $B_0 = (L_0^{\mathsf{T}} L_0)^{-1}$, $N_0 = JL_0B_0$, *and* $W_0 = D_\nu L_0$. *Let* $l_0, l_0^t, b_0, n_0, n_0^t, w_0, w_0^t, h_0, z_0$ *be valid bounds for* $L_0, L_0^{\mathsf{T}}, B_0, N_0, N_0^{\mathsf{T}}, W_0, W_0^{\mathsf{T}}, D_\nu(L_0^{\mathsf{T}} L_0), D_\nu N_0$. *Write*

$$g = l_0^t d + d^t l_0 + d^t d, \quad \eta_B = b_0 g.$$

If $\eta_B < 1$, *the varying Gram matrix is invertible and*

$$b = \frac{b_0}{1-\eta_B}, \quad \Delta b = \frac{b_0^2 g}{1-\eta_B}, \quad \Delta n = db + l_0 \Delta b, \quad \Delta n^t = \Delta b\, l_0^t + b d^t$$

bound B, $B - B_0$, $N - N_0$ *and* $N^{\mathsf{T}} - N_0^{\mathsf{T}}$. *One may take* $n = \min\{(l_0 + d)b, n_0 + \Delta n\}$ *and* $n^t = \min\{b(l_0^t + d^t), n_0^t + \Delta n^t\}$.

Set

$$\begin{aligned} \Delta h &= w^t l_0 + w_0^t d + w^t d + d^t w_0 + l_0^t w + d^t w, \\ h &= h_0 + \Delta h, \quad c = b^2 h, \\ \Delta c &= \Delta b\, hb + b_0 \Delta h\, b + b_0 h_0 \Delta b, \\ \Delta z &= wb + w_0 \Delta b + dc + l_0 \Delta c. \end{aligned}$$

Then $D_\nu N$ *is bounded by* $z = z_0 + \Delta z$. *For the saved canonical torsion convention*

$$T_\lambda(P) = N^{\mathsf{T}} J\{DX_\lambda(P)N - D_\nu N\},$$

define

$$q_0 = c_1 n_0 + z_0, \quad \Delta q = c_2 v n + c_1 \Delta n + \Delta z, \qquad t = \Delta n^t (q_0 + \Delta q) + n_0^t \Delta q.$$

Provided all straight state segments remain in the common convex analytic rectangle, uniformly in real λ,

$$\|\langle T_\lambda(P_\lambda)\rangle - \langle T_\lambda(K_0)\rangle\|_\infty \le t.$$

Proof. Expand the Gram difference and use the inverse Neumann identity. The normal-frame bounds follow from $N = JLB$ and analytic transpose symmetry $B^{\mathsf{T}} = B$. For the directional derivative use the exact identities $C = D_\nu B = -B(W^{\mathsf{T}} L + L^{\mathsf{T}} W)B$ and $D_\nu N = JWB + JLC$. The displayed Δh retains all six terms of the middle-factor difference; telescoping the two outer inverse factors gives Δc. Next, $\|DX_\lambda(P) - DX_\lambda(K_0)\| \le c_2 v$. Expand $DX_\lambda(P)N - D_\nu N$ about K_0, then expand its left N^{T} factor. Averaging cannot increase this entrywise analytic norm. No exact invariance or Lagrangian identity is used.

The accepted signed fixed-map certificate supplies $I_* \ge \sup_{\lambda\in[0,1]} \|\langle T_\lambda(K_0)\rangle^{-1}\|_\infty$. Consequently $I_* t < 1$ is a sufficient varying-map nondegeneracy test, with inverse bound $I_*/(1 - I_* t)$. Its failure is not a singularity certificate. The current $I_* \simeq 19036.55$ requires a change less than approximately $5.253\,10^{-5}$ for this particular sufficient test. An unsigned pointwise transport can be much larger than the actual signed continuous mean.

A genuine direct frame inverse. The construction frame is $M = [L, JLB]$, not an assumed exactly symplectic frame. Put $\Omega = L^{\mathsf{T}} JL$. Since B is symmetric,

$$M^{\mathsf{T}} JM = \begin{pmatrix} \Omega & -I \\ I & B\Omega B \end{pmatrix}.$$

Thus $\kappa = \max(1, b^2)\|\Omega\| < 1$ proves M invertible. It gives

$$\|M^{-1}\| \le \frac{\max(n^t, l^t)}{1-\kappa}, \qquad \left\| M^{-1} - \begin{pmatrix} N^{\mathsf{T}} J \\ -L^{\mathsf{T}} J \end{pmatrix} \right\| \le \frac{\kappa \max(n^t, l^t)}{1-\kappa}.$$

The implementation bounds Ω_0 from the actual harmonic normalization S: it computes the exact rational $\|S^{\mathsf{T}}JS - J\|$, adds the normalized DK_0 remainder products, and then adds the varying DK products. It does not assume that the stored floating-point S is exactly symplectic. Direct-frame reduced upper-right torsion is used to construct responses. The KAM theorem's saved torsion above is supplied separately; these two matrices are not declared equal at a non-Lagrangian approximate embedding.

The eight actual scalar gates. Use the unchanged certified (ν, γ, τ, c_R), $\rho = 1/40$, $\rho_\infty = 1/400$, $a_1 = 2$, and the ordinary canonical $n = d = 2$ Haro–Luque Case III constants. Replace the scalar calculator inputs by the newly proved uniform

$$k = \sup \|DP_\lambda\|, \quad k^t = \sup \|(DP_\lambda)^{\mathsf{T}}\|, \quad b = \sup \|B_\lambda\|, \quad I = \sup \|\langle T_\lambda \rangle^{-1}\|,$$

the remaining domain radius and (c_0, c_1, c_{t1}, c_2) above. Choose strict auxiliary bounds $\sigma_K > k$, $\sigma_{K^t} > k^t$, $\sigma_B > b$, $\sigma_T > I$; the supplied finite calculator uses the admissible ratio $11/10$. All Table 1–3 constants are recomputed from these inputs by the same previously audited exact-rational constants engine. This is shared constants code, not a new independent derivation of all table entries.

To make the mapping explicit, write $\ell = \rho/a_3$ for its strip loss, $E = \sup_{\lambda\in[0,1]} \|X_\lambda(P_\lambda) - D_\nu P_\lambda\|_\rho$, and denote its newly computed table constants by $C_E, C_{\mathrm{sym}}, C_K, C_B, C_{T^{-1}}$. The required dimensionless inequalities, each strictly less than one, are

$$\begin{aligned}
g_1 &= (a_1 a_3)^{4\tau} C_E E/(\gamma^4 \rho^{4\tau}),\\
g_2 &= E/(\ell \nu_{\mathrm{iter}}), \qquad g_3 = 2C_{\mathrm{sym}} E/(\gamma \ell^{\tau+1}),\\
g_4 &= \frac{2C_K E}{(\sigma_K - k)\gamma^2 \ell^{2\tau+1}(1 - a_1^{1-2\tau})},\\
g_5 &= \frac{4C_K E}{(\sigma_{K^t} - k^t)\gamma^2 \ell^{2\tau+1}(1 - a_1^{1-2\tau})},\\
g_6 &= \frac{C_B E}{(\sigma_B - b)\gamma^2 \ell^{2\tau+1}(1 - a_1^{1-2\tau})},\\
g_7 &= \frac{C_{T^{-1}} E}{(\sigma_T - I)\gamma^2 \ell^{2\tau+1}(1 - a_1^{1-2\tau})},\\
g_8 &= \frac{C_K E}{r_{\mathrm{dom}}\gamma^2 \ell^{2\tau}(1 - a_1^{-2\tau})}.
\end{aligned}$$

These are respectively contraction, residual/iteration, approximate symplectic inverse, preservation of DP, its transpose, Gram inverse, mean torsion inverse and domain distance. Here $\nu_{\mathrm{iter}} = 1$ is a theorem parameter and is not the two-component frequency ν. The calculator evaluates the eight slopes g_i/E with exact rational arithmetic; their reciprocal maximum is a new conditional residual threshold. If I is only a proposed target, every resulting gate and threshold remains explicitly conditional. A scalar calculation does not turn that target into a proved mean-inverse bound.

The KAM C^0 correction is its newly computed $C_2 E/(\gamma^2 \rho^{2\tau})$. For real embedding use the actual retained-strip normalized harmonic errors, adding $\|S^{-1}\|\|V\|$ and $\|S^{-1}\|\|DV\|$. Then add the KAM correction to the former and $2\|S^{-1}\|$ times that correction divided by ρ_∞ to the latter. The already audited normalized phase test

$$a > 2e_0, \qquad \frac{2e_1 + (88/7)e_0}{6(a - 2e_0)} < \tfrac{1}{2}$$

must hold uniformly. The preliminary response-only phase test in the finite calculator does not include an as-yet unknown KAM correction.

Proposition S8. Conditional uniform family acceptance. *If the full same-chain residual E, uniform domain, derivatives, Gram inverse and continuous mean inverse are verified as stated, all eight fresh gates pass, and the corrected normalized phase test passes, then the canonical a-posteriori theorem applies for each real $\lambda \in [0, 1]$ to this actual four-coordinate pair, with the same prescribed frequency. In particular it applies to the physical endpoint. No analytic exact K_λ family is assumed in constructing P_m.*

The lower residuals remain actual nonzero coefficients. In particular $\langle DK_0^{\mathsf{T}} J e_j\rangle = -\sum_{r=1}^{j}\langle DU_r^{\mathsf{T}} J e_{j-r}\rangle$. This compatibility identity and the approximate-symplectic gate replace any invalid assumption of exact Lagrangianity. A passed learned-model threshold, the fixed-map signed inverse, or the newly verified e_1 alone is not a substitute for the full varying-family hypotheses.

A concrete alternative when unsigned torsion transport fails. Use the actual P_m on a containing angle–parameter product domain and certify its Gram inverse there. Evaluate the parameter Taylor coefficients of $\langle T_\lambda(P_m)\rangle$, retaining their signed two-by-two entries and complete angle-quadrature aliases. A modulus M_T on $|\lambda| \le R$ bounds the tail after L_T by $M_T/[R^{L_T}(R-1)]$. Verify the determinant is separated from zero on real $[0,1]$, for example by exact interval Bernstein subdivision of the signed matrix polynomial with the entrywise tail added, and bound its inverse by adjugate/determinant. No symmetry of the approximate saved mean is assumed. The necessary series graph includes physical G, G_q, G_{qq}, the actual network Hessian, the derivative of $N = JLB$ and its analytic Gram inverse. This is a specific new continuous-mean certificate to compute; it cannot be inherited from the fixed K_0.

S6 Observation-based acceptance of a full-phase Hamiltonian family

Let $x = (q_1, q_2, p_1, p_2)$ be physical canonical coordinates, let $J = \left(\begin{smallmatrix} 0 & I_2 \\ -I_2 & 0 \end{smallmatrix}\right)$, and fix a prescribed reference and a learned energy

$$H_* = T_G + V_4 + U(q), \qquad \widehat{H}_\vartheta = T_G + V_4 + N_\vartheta(q,p), \qquad T_G = \tfrac{1}{2} p^{\mathsf{T}} G(q)^{-1} p. \tag{S11}$$

The real scalar N_ϑ may be supplied by a complex-valued network with a real, coefficient-sharp readout. Its complexification must be holomorphic on every domain used below. No holomorphicity in the physical variable $q + ip$ is imposed, and no momentum derivative of N_ϑ is set to zero. Define

$$e_\vartheta = N_\vartheta - U, \quad Y_\vartheta = -J\nabla e_\vartheta, \quad H_{\vartheta,\lambda} = T_G + V_4 + (1-\lambda)N_\vartheta + \lambda U, \qquad 0 \le \lambda \le 1. \tag{S12}$$

Thus $X_{H_{\vartheta,\lambda}} = X_{\widehat{H}_\vartheta} + \lambda Y_\vartheta$.

A finite observation certificate. Fix positive coordinate scales s_j and the complete set χ of nonconstant even-total-degree monomials in x_j/s_j through degree $D = 4$ or 6. Its dimension is 45 or 129. For an output choose an identified polynomial $P_\vartheta = \chi^{\mathsf{T}} a_\vartheta$ and write $\rho_\vartheta = N_\vartheta - P_\vartheta$. This decomposition does not restrict the network to a polynomial class.

The observation operator $\mathcal{L}$ contains the selected physical configuration derivatives at rest anchors $(Q_i, 0)$ and momentum derivatives at moving anchors. The latter may use the exact observation chart $(\bar{q}, G(\bar{q})\bar{v})$. The reference correction satisfies $U_p = 0$ at every such anchor. This is a consequence of the shared kinetic graph in (S11), independent of any assertion that the chart point is a true trajectory state. At rest, labels have the form $Y_i = \nabla(V_4 + U)(Q_i) + \xi_i$. With the selected components, the actual residual is

$$r_{q,i} = N_{\vartheta,q}(Q_i, 0) + V_{4,q}(Q_i) - Y_i, \qquad r_{p,l} = N_{\vartheta,p}(\bar{q}_l, G(\bar{q}_l)\bar{v}_l).$$

Here G and V_4 are the same prescribed scalar graph as in (S11). The resulting signed residual obeys

$$r_\vartheta = \mathcal{L} e_\vartheta - \xi. \tag{S13}$$

All label, state, clock, discretization and numerical allowances needed for this identity must be included in a common fidelity certificate. When nominal and true anchors differ, their joint interval enclosure or a sound state-transport allowance is required. Evaluation widths are retained separately from physical label errors.

Let W be a declared positive diagonal row weight, applied to the residual, label errors and observation remainders alike. Put $\widetilde{\mathcal{L}} = W\mathcal{L}$, $A = \widetilde{\mathcal{L}}\chi^{\mathsf{T}}$ and $\widetilde{r}_\vartheta = W r_\vartheta$. Each physical derivative includes its factor $1/s_j$. The observation design and its inverse are independent of the candidate, although they may depend on the observed input geometry. Every candidate's model and remainder certificates refer to this same declared design. For the square design suppose a fixed exact matrix R satisfies

$$\sup_A \|I - RA\|_\infty \le \eta < 1, \qquad J_A = (RA)^{-1}R, \qquad \kappa = \eta/(1-\eta), \tag{S14}$$

where the supremum covers the actual joint anchor enclosure. Column permutations, row order and weights are common to all terms. Write $F_\alpha(x) = \partial_x^\alpha \chi(x)^\mathsf{T}$ and define

$$C_{\alpha,\vartheta}(x) = \partial_x^\alpha(\rho_\vartheta - U)(x) - F_\alpha(x) J_A \widetilde{\mathcal{L}}(\rho_\vartheta - U). \tag{S15}$$

Lemma S9. Pathwise output-error bounds. *Let Ω be a real or complex target rectangle, with valid modulus bounds $|x_j| \le r_j$. Define*

$$g_{\alpha,\beta} = \prod_{j=1}^{4} \frac{(\beta_j)_{\alpha_j} r_j^{\beta_j - \alpha_j}}{s_j^{\beta_j}}, \quad G_\alpha = \sum_\beta g_{\alpha,\beta}, \qquad \mathcal{G}_\alpha(v) = \sum_\beta g_{\alpha,\beta} v_\beta + G_\alpha \kappa \|v\|_\infty,$$

with $g_{\alpha,\beta} = 0$ if some $\beta_j < \alpha_j$. Suppose $|W\xi| \le \zeta$ and $c_{\alpha,\vartheta} \ge \sup_\Omega |C_{\alpha,\vartheta}|$ are sound. Then

$$\|\partial_x^\alpha e_\vartheta\|_\Omega \le b_{\alpha,\vartheta} := \mathcal{G}_\alpha(|R\widetilde{r}_\vartheta|) + \mathcal{G}_\alpha(|R|\zeta) + c_{\alpha,\vartheta}. \tag{S16}$$

The signed matrix–vector product in the first term is enclosed before taking absolute values. On a common fidelity event this conclusion holds simultaneously for every output having sound individual certificates, including an output selected from the same observations.

Proof. The Neumann series in (S14) gives $J_A A = I$ and $\|(RA)^{-1} - I\|_\infty \le \kappa$. By (S13),

$$A a_\vartheta = \widetilde{r}_\vartheta + W\xi - \widetilde{\mathcal{L}}(\rho_\vartheta - U).$$

Substitution into the differentiated error decomposition gives the exact identity

$$\partial_x^\alpha e_\vartheta = F_\alpha J_A \widetilde{r}_\vartheta + F_\alpha J_A W\xi + C_{\alpha,\vartheta}.$$

For any vector y, write $J_A y = Ry + [(RA)^{-1} - I]Ry$. The monomial envelope bounds the first term by $\sum_\beta g_{\alpha,\beta}|(Ry)_\beta|$ and the second by $G_\alpha\kappa\|Ry\|_\infty$. Use $|RW\xi| \le |R|\zeta$ and the remainder bound to obtain (S16). The argument is deterministic for each data realization and uses no selection rule for ϑ. Both the observation-point remainder and the target-domain remainder remain in (S15); neither is discarded.

A prospective independent-anchor alternative. Split χ into configuration-only columns χ_q and mixed columns χ_p. Their dimensions are $(d_q, d_p) = (8, 37)$ or $(15, 114)$. Write $x_j = s_j u_j$ with fixed $s_j > 0$, and put $D_q = \mathrm{diag}(s_1, s_2)$ and $D_p = \mathrm{diag}(s_3, s_4)$. For $b = q, p$, let

$$B_b(u) = D_b \nabla_{x,b} \chi_b(x(u))^\mathsf{T}, \qquad A_b^{(i)} = \pi_b(u_i)^{-1/2} B_b(u_i),$$

where $\nabla_{x,q} = (\partial_{q_1}, \partial_{q_2})^\mathsf{T}$ at rest and $\nabla_{x,p} = (\partial_{p_1}, \partial_{p_2})^\mathsf{T}$ at moving canonical anchors. Thus B_b contains derivatives with respect to the normalized coordinates on $[-1,1]^2$ or $[-1,1]^4$, respectively, and A_b stacks the displayed two-row blocks. Put

$$M_b = \mathbb{E}_{\mathrm{unif}} B_b^\mathsf{T} B_b, \qquad K_b(u) = \mathrm{tr}(M_b^{-1} B_b(u)^\mathsf{T} B_b(u)).$$

The column sets exclude the respective derivative nullspaces, so $M_b \succ 0$. Fix the scales and densities $\pi_b > 0$ relative to uniform probability before drawing the anchors. Draw n_b anchors independently within each block and use row weights $\pi_b(u)^{-1/2}$. Suppose $K_b/\pi_b \le L_b$ almost everywhere. For $0 < \varepsilon_b < 1$ and $0 < \alpha_b < 1$, the sufficient count

$$n_b \ge \frac{2L_b}{\varepsilon_b^2} \log \frac{d_b}{\alpha_b} \tag{S17}$$

gives a common event $\mathcal{E}_{\mathrm{des}}$, of probability at least $1 - \alpha_q - \alpha_p$, on which

$$n_b^{-1} A_b^\mathsf{T} A_b \succeq (1 - \varepsilon_b) M_b, \qquad b = q, p. \tag{S18}$$

Each anchor contributes one positive semidefinite two-row product; the two derivative rows at an anchor need not be independent. The representation, scales, matrices, leverage functions and laws are fixed before collection. For example, $\pi_b = (1 + K_b/d_b)/2$ is a normalized full-support density and permits $L_b = 2d_b$.

Write the actual error as $e_\vartheta = \chi_q^{\mathsf{T}} c_{q,\vartheta} + \chi_p^{\mathsf{T}} c_{p,\vartheta} + w_\vartheta$. Any reference polynomial in this identity is used for verification, not as a prescribed training label. For each anchor, transform the physical residual, fidelity error and remainder gradient by the same diagonal scale and importance weight:

$$r_{b,\vartheta}^{(i)} = \pi_b(u_i)^{-1/2} D_b r_{b,\vartheta}^{\mathrm{phys,(i)}}, \qquad \xi_b^{(i)} = \pi_b(u_i)^{-1/2} D_b \xi_b^{\mathrm{phys,(i)}},$$
$$(\mathcal{L}_b w_\vartheta)^{(i)} = \pi_b(u_i)^{-1/2} D_b \nabla_{x,b} w_\vartheta(x(u_i)).$$

Here the physical block residual has the sign convention $r_{b,\vartheta}^{\mathrm{phys,(i)}} = \nabla_{x,b} e_\vartheta(x(u_i)) - \xi_b^{\mathrm{phys,(i)}}$. The stacked block residual identity is therefore

$$r_{b,\vartheta} = A_b c_{b,\vartheta} + \mathcal{L}_b w_\vartheta - \xi_b.$$

With every term scaled and weighted as above, let

$$R_{b,\vartheta} = \|r_{b,\vartheta}\|_2/\sqrt{n_b}, \quad Z_b \geq \|\xi_b\|_2/\sqrt{n_b}, \quad T_{b,\vartheta} \geq \|\mathcal{L}_b w_\vartheta\|_2/\sqrt{n_b}.$$

If

$$C_{\alpha,b} \geq \sup_{x\in\Omega} (F_{\alpha,b} M_b^{-1} F_{\alpha,b}^*)^{1/2}, \qquad W_{\alpha,\vartheta} \geq \sup_{\Omega} |\partial_x^\alpha w_\vartheta|,$$

then a second admissible budget is

$$b_{\alpha,\vartheta} = \sum_{b=q,p} C_{\alpha,b} \frac{R_{b,\vartheta} + Z_b + T_{b,\vartheta}}{\sqrt{1-\varepsilon_b}} + W_{\alpha,\vartheta}. \tag{S19}$$

Here $*$ conjugates the target-domain row. Counts in (S17) count independent anchors, not derivative rows or dependent trajectory snapshots.

Lemma S10. Common design and fidelity events. *The design assumptions above imply* (S18) *with the stated probability. On its intersection with a common event $\mathcal{E}_{\mathrm{fid}}$ supporting the fidelity allowances, the derivative bounds* (S19) *hold simultaneously for every soundly certified output. No independence between the two block events or between design and fidelity events is needed for the union bound.*

Proof. Positive definiteness follows because a vanishing rest gradient makes a configuration polynomial constant, whereas a vanishing momentum gradient makes a mixed polynomial configuration-only. The specified columns exclude both nullspaces. Moreover $\mathbb{E}K_b = d_b$, so the displayed mixture density integrates to one, is at least $1/2$, and satisfies $K_b/\pi_b = 2d_bK_b/(d_b + K_b) \leq 2d_b$. For one block, the independent matrices

$$M_b^{-1/2} B_b(u_i)^{\mathsf{T}} B_b(u_i) M_b^{-1/2}/\pi_b(u_i)$$

have expectation I, are positive semidefinite and have largest eigenvalue at most their trace L_b. The lower matrix Chernoff estimate of Lemma S4 bounds failure by $d_b \exp[-\varepsilon_b^2 n_b/(2L_b)] \leq \alpha_b$. The union bound gives the joint event. On that event the residual identity and triangle inequality imply

$$\|M_b^{1/2} c_{b,\vartheta}\|_2 \leq (R_{b,\vartheta} + Z_b + T_{b,\vartheta})/\sqrt{1-\varepsilon_b}.$$

Complex Cauchy–Schwarz applied to $F_{\alpha,b} M_b^{-1/2} M_b^{1/2} c_{b,\vartheta}$ proves (S19). Each matrix event holds for all coefficient vectors, whereas each output's actual remainder must be certified. Consequently selection of that output using the observations does not require a union bound over network parameters.

Uniform hypotheses for the conservative family. Use either of the preceding derivative budgets for $1 \le |\alpha| \le 3$. Define

$$\delta_0 = \max_i b_{\mathbf{e}_i,\vartheta}, \qquad \delta_1 = \max_i \sum_j b_{\mathbf{e}_i+\mathbf{e}_j,\vartheta},$$
$$\delta_{t1} = \sum_{i,j} b_{\mathbf{e}_i+\mathbf{e}_j,\vartheta}, \qquad \delta_2 = \max_i \sum_{j,k} b_{\mathbf{e}_i+\mathbf{e}_j+\mathbf{e}_k,\vartheta}. \tag{S20}$$

These bound Y_ϑ, DY_ϑ, its transpose and its ordered second derivative. If the actual learned field has bounds $(\widehat{c}_0, \widehat{c}_1, \widehat{c}_{t1}, \widehat{c}_2)$ on Ω, valid uniform real-parameter inputs are

$$(c_0, c_1, c_{t1}, c_2) = (\widehat{c}_0 + \delta_0, \widehat{c}_1 + \delta_1, \widehat{c}_{t1} + \delta_{t1}, \widehat{c}_2 + \delta_2). \tag{S21}$$

Every sum retains physical-coordinate factors and ordered multiplicities.

For a prescribed period-one frequency ν, consider the same output's exact finite embedding polynomial

$$P_{\vartheta,m}(\varphi, \lambda) = K_{\vartheta,0}(\varphi) + \sum_{j=1}^{m} \lambda^j U_{\vartheta,j}(\varphi), \qquad D_\nu = \nu \cdot \partial_\varphi.$$

Each response and every subsequent certificate refer to its complete preceding prefix. Let $\|\cdot\|_\rho$ denote the componentwise maximum analytic norm on the angle strip of half-width ρ; for a matrix use $\|A\|_\rho = \max_i \sum_j \sup_{\mathbb{T}^2_\rho} |A_{ij}|$, with transpose bounds supplied separately. Impose the following quantitative hypotheses. Fix retained widths $0 < \rho_\infty < \rho$.

1. The Hamiltonians in (S12) are real on real states and analytic on a common domain. For real $\lambda \in [0,1]$, the image $P_{\vartheta,m}(\mathbb{T}^2_\rho, \lambda)$ has a strict uniform tube of radius $r_{\mathrm{dom}} > 0$ in Ω. The metric is invertible there, and (S21) bounds the actual field and its derivatives on this tube.
2. The frequency satisfies $|k \cdot \nu| \ge \gamma |k|_1^{-\tau}$ for $k \in \mathbb{Z}^2 \setminus \{0\}$, with certified $\gamma > 0$, $\tau \ge 1$ and the corresponding period-one cohomological constant. For every real λ, write

   $$L_\lambda = D_\varphi P_{\vartheta,m}, \quad B_\lambda = (L_\lambda^{\mathsf{T}} L_\lambda)^{-1}, \quad \mathcal{N}_\lambda = J L_\lambda B_\lambda.$$

   The Gram inverse is defined on the whole strip, and there are uniform bounds $k \ge \|L_\lambda\|_\rho$, $k^t \ge \|L_\lambda^{\mathsf{T}}\|_\rho$, $b \ge \|B_\lambda\|_\rho$ and

   $$I \ge \sup_{\lambda\in[0,1]} \|\langle T_\lambda \rangle^{-1}\|_\infty, \qquad T_\lambda = \mathcal{N}_\lambda^{\mathsf{T}} J \{DX_{H_{\vartheta,\lambda}}(P_{\vartheta,m})\mathcal{N}_\lambda - D_\nu \mathcal{N}_\lambda\}. \tag{S22}$$

   The average is that of this varying embedding, with its signed entries and complete integration errors. Transposes are analytic transposes. Neither exact Lagrangianity nor equality with a construction frame's reduced torsion is assumed for the approximate embedding.
3. The full defect

   $$\mathcal{R}_\vartheta(\varphi, \lambda) = X_{H_{\vartheta,\lambda}}(P_{\vartheta,m}) - D_\nu P_{\vartheta,m}$$

   has a certified bound $E \ge \sup_{\lambda\in[0,1]} \|\mathcal{R}_\vartheta\|_\rho$. In particular, if it is analytic on $\mathbb{T}^2_{\widehat{\rho}} \times \{|\lambda| \le R_\lambda\}$ with $\widehat{\rho} > \rho$, $R_\lambda > 1$ and modulus at most M, write $\mathcal{R}_\vartheta = \sum_{j\ge 0} e_{\vartheta,j} \lambda^j$. For any finite set $S \subseteq \{0, \dots, L\}$ and sound $\epsilon_j \ge \|e_{\vartheta,j}\|_\rho$, a complete admissible bound is

   $$E = \sum_{j\in S} \epsilon_j + \sum_{\substack{0\le j\le L \\ j\notin S}} M/R_\lambda^j + \frac{M}{R_\lambda^L (R_\lambda - 1)}. \tag{S23}$$

   Numerical Fourier coefficients retain their interval radii, all representative modes including Nyquist, and all infinite angle aliases. For an even grid size n, the complete interpolation allowance for coefficient j is

   $$2\frac{M}{R_\lambda^j}\left(\frac{1+q}{1-q}\right)^2 (2q^{n/2} - q^n), \qquad q = e^{-2\pi(\widehat{\rho}-\rho)}.$$

Any symmetry reducing the sampling is an exact identity of the specified field and embedding. The outer product domain is certified separately; the bounds on Ω do not imply its enlargement to that domain.

4. Choose admissible strip-loss and iteration parameters of the canonical analytic a posteriori theorem, including $\ell = \rho/a_3$, $a_1 > 1$, and a positive iteration parameter ν_{iter}. From the uniform inputs just listed, recompute its Case III constants $C_E, C_{\mathrm{sym}}, C_K, C_B, C_{T^{-1}}, C_2$. Choose $\sigma_K > k$, $\sigma_{K^t} > k^t$, $\sigma_B > b$ and $\sigma_T > I$. All eight dimensionless tests must be strictly below one:

$$
\begin{aligned}
g_1 &= \frac{(a_1 a_3)^{4\tau} C_E E}{\gamma^4 \rho^{4\tau}}, & g_2 &= \frac{E}{\ell \nu_{\mathrm{iter}}}, & g_3 &= \frac{2C_{\mathrm{sym}} E}{\gamma \ell^{\tau+1}},\\
g_4 &= \frac{2C_K E}{(\sigma_K - k)\gamma^2 \ell^{2\tau+1}(1 - a_1^{1-2\tau})}, & g_5 &= \frac{4C_K E}{(\sigma_{K^t} - k^t)\gamma^2 \ell^{2\tau+1}(1 - a_1^{1-2\tau})},\\
g_6 &= \frac{C_B E}{(\sigma_B - b)\gamma^2 \ell^{2\tau+1}(1 - a_1^{1-2\tau})}, & g_7 &= \frac{C_{T^{-1}} E}{(\sigma_T - I)\gamma^2 \ell^{2\tau+1}(1 - a_1^{1-2\tau})},\\
g_8 &= \frac{C_K E}{r_{\mathrm{dom}} \gamma^2 \ell^{2\tau}(1 - a_1^{-2\tau})}.
\end{aligned}
\tag{S24}
$$

The constants and auxiliary parameters have the conventions of Section S5; they belong to these uniform inputs, rather than to a different Hamiltonian or embedding.

5. The input real embeddings and the theorem's corrected real embeddings satisfy a sound uniform embedding test, including the C^0 correction $\Delta = C_2 E/(\gamma^2 \rho^{2\tau})$ and its retained-strip derivative allowance. This may be verified by the normalized harmonic phase criterion with the correction added to both norm budgets.

Theorem S11. Learning-to-family acceptance on a common event. *Suppose a common fidelity event $\mathcal{E}_{\mathrm{fid}}$ has probability at least $1 - \alpha_{\mathrm{fid}}$. On this event let either the strict finite observation certificate of Lemma S9 apply, or let the independent-anchor construction satisfy (S17) and Lemma S10. An output is accepted only when its own sound derivative, remainder and domain certificates give (S20) and all five uniform hypotheses above are verified for that same output, reference and complete embedding prefix.*

On the corresponding common good event, every accepted output has the certified derivative bounds and, for every real $\lambda \in [0, 1]$, a real analytic invariant torus $K^*_{\vartheta,\lambda}$ with the prescribed frequency:

$$
X_{H_{\vartheta,\lambda}}(K^*_{\vartheta,\lambda}) = D_\nu K^*_{\vartheta,\lambda}, \qquad \|K^*_{\vartheta,\lambda} - P_{\vartheta,m}(\cdot, \lambda)\|_{\rho_\infty} \le \Delta. \tag{S25}
$$

The constant Δ is uniform in λ. In particular $\lambda = 1$ gives an invariant torus of the prescribed conservative reference H_*.

For any measurable data-dependent rule that returns an output or rejects, including one using the same observations for training and selection, let A *denote acceptance and* F *the event that a stated derivative or conservative-family conclusion for the returned output is false. Then*

$$
\mathbb{P}(\mathsf{A} \cap \mathsf{F}) \le \begin{cases} \alpha_{\mathrm{fid}}, & \textit{strict finite observation route},\\ \alpha_{\mathrm{fid}} + \alpha_q + \alpha_p, & \textit{independent-anchor route}. \end{cases} \tag{S26}
$$

If all fidelity allowances are deterministic, the first route is a deterministic implication. The theorem supplies neither a lower bound on acceptance probability nor a guarantee that optimization produces an acceptable output. It makes no conditional-confidence assertion given acceptance.

Proof. Fix a realization in the common good event and any accepted output. Lemma S9, or Lemma S10, bounds every required derivative of the actual scalar e_ϑ. Canonical multiplication by J only permutes coordinates and changes signs. The triangle inequality and ordered derivative sums therefore give (S20). For real $0 \le \lambda \le 1$, the identity $X_{H_{\vartheta,\lambda}} = X_{\widehat{H}_\vartheta} + \lambda Y_\vartheta$ then proves (S21) on the certified inner tube.

For completeness, Cauchy's estimate on the separate product domain gives $\|e_{\vartheta,j}\|_{\widehat{\rho}} \le M/R_\lambda^j$. The triangle inequality on $|\lambda| \le 1$, followed by summation of the infinite geometric tail, proves (S23). The Fourier interpolation allowance adds all omitted modes and their aliases, while interval sample and transform widths remain in the finite interpolant. No coefficient above the input polynomial degree is set to zero in this argument.

For each fixed real λ, the certified common domain, nonresonance, Gram and actual mean inverse bounds satisfy the analytic canonical a posteriori theorem's hypotheses. Its constants are bounded by the recomputed uniform constants in (S24); the eight strict inequalities provide its contraction and preservation conditions. That theorem yields a real analytic invariant map with the bound Δ. The final embedding test gives the asserted real torus embedding. Applying this deterministic implication separately to each real λ proves (S25) with the same bound. Analytic dependence of the exact tori on λ is not needed for this statement.

All implications just established hold simultaneously for every accepted output on the common event. A false accepted conclusion is therefore contained in its complement. The common fidelity bound, and in the independent-anchor case the union bound of Lemma S10, prove (S26). No union over possible selected network parameters is required.

The independent-anchor assertion requires its specified sampling law and common fidelity event. A collection of dependent trajectories does not acquire that law by contributing many time nodes. A successful conditioning event supplies the coefficient inequality; small observed residuals and actual remainders are additional acceptance requirements. These conclusions concern the specified finite-dimensional reference equations and are neutral with respect to comparisons between real and complex parametrizations.

Proposition S12. A separate dissipative port structure. *On a real domain $\mathcal{D} \subset \Omega \cap \mathbb{R}^4$ with $G = SS^{\mathsf{T}} \succ 0$, let a configuration-only complex head define*

$$\widehat{M}_\psi(q) = \mathrm{Re}(L_\psi(q)L_\psi(q)^*), \quad \widehat{R}_\psi = S\widehat{M}_\psi S^{\mathsf{T}}, \quad \widehat{v} = \partial_p \widehat{H}_\vartheta = G^{-1}p + N_{\vartheta,p}.$$

For the specified magnetic port $g(q) = d\cos(d^{\mathsf{T}}q)$ and the learned port-Hamiltonian equations

$$q' = \widehat{v}, \qquad p' = -\partial_q \widehat{H}_\vartheta - \widehat{R}_\psi \widehat{v} + g(q)u,$$

the head is real symmetric positive semidefinite and

$$\frac{d\widehat{H}_\vartheta}{dt} = -\widehat{v}^{\mathsf{T}}\widehat{R}_\psi\widehat{v} + u\, g^{\mathsf{T}}\widehat{v}. \tag{S27}$$

For the reference plant with $v_* = G^{-1}p$ and $R_* = SM_*S^{\mathsf{T}} \succeq 0$, set $e_p = N_{\vartheta,p} = \widehat{v} - v_*$. If $\|\widehat{M}_\psi - M_*\|_2 \le \delta_M$ is separately certified, and $a \ge \|S^{\mathsf{T}}\widehat{v}\|_2$, $b \ge \|S^{\mathsf{T}}e_p\|_2$ and $\bar{m} \ge \|\widehat{M}_\psi\|_2$, then

$$\begin{aligned} |g^{\mathsf{T}}v_* - g^{\mathsf{T}}\widehat{v}| &\le \sum_j |g_j|\, b_{\mathbf{e}_{2+j},\vartheta}, \\ v_*^{\mathsf{T}}R_*v_* &\ge \widehat{v}^{\mathsf{T}}\widehat{R}_\psi\widehat{v} - 2\bar{m}ab - \delta_M(a+b)^2. \end{aligned} \tag{S28}$$

These port identities do not apply the conservative KAM theorem to a dissipative system. Reference-plant safety additionally requires the energy, state enclosure, guard, command and intersample hypotheses of the physical-energy safety theorem.

Proof. For a real vector z, $z^{\mathsf{T}}\widehat{M}_\psi z = \|L_\psi^* z\|_2^2 \ge 0$. The two Hamiltonian terms cancel by skew symmetry, proving (S27) with the full learned velocity. Since $U_p = 0$, the learned-to-reference velocity error is exactly $e_p = \partial_p e_\vartheta = N_{\vartheta,p}$. The first inequality of (S28) follows from its componentwise derivative bounds. For the second put $z = S^{\mathsf{T}}\widehat{v}$ and $\eta = S^{\mathsf{T}}e_p$. Expand $(z-\eta)^{\mathsf{T}}M_*(z-\eta)$ about $\widehat{M}_\psi$, retain its nonnegative quadratic term in η, and bound the cross term and matrix error by $2\bar{m}ab$ and $\delta_M(a+b)^2$. All momentum-error terms are retained. The reference physical-power identity is $H_*' = u\, g^{\mathsf{T}}v_* - v_*^{\mathsf{T}}R_*v_*$; its sampled safety implications require the separate stated hypotheses.

Canonical conventions for the conservative continuation. The following identifies the canonical hypotheses used in Theorem S11 with those of Haro–Luque, arXiv:1711.05515v1, Theorem 3.1 and Remark 3.3. Put

$$J_s = \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix}, \qquad \Omega_{\mathrm{HL}} = -J_s, \qquad g_{\mathrm{amb}}(u,v) = u^{\mathsf{T}}v.$$

The symplectic form is exact, with primitive $\sum_{j=1}^{2} p_j\,dq_j$, and the Hamiltonian convention is $\Omega_{\mathrm{HL}}X_H = (DH)^{\mathsf{T}}$, hence $X_H = J_s\nabla H$. In the canonical compatible Case III, $J_{\mathrm{HL}} = \Omega_{\mathrm{HL}} = -J_s$ and the auxiliary metric matrix is I_4. This auxiliary metric is independent of the mechanical inertia $G_8(q)$ in $\frac{1}{2}p^{\mathsf{T}}G_8(q)^{-1}p$.

For a fixed real parameter λ, write $L = D_\varphi P_m(\varphi,\lambda)$ and $B = (L^{\mathsf{T}}L)^{-1}$, using analytic transposes. The saved normal and torsion conventions are

$$N_s = J_sLB, \qquad T_s = N_s^{\mathsf{T}}J_s\{DX_{H_\lambda}(P_m)N_s - D_\nu N_s\}.$$

The theorem's normal satisfies $N_{\mathrm{HL}} = -N_s$. Its torsion is therefore

$$\begin{aligned} T_{\mathrm{HL}} &= (-N_s)^{\mathsf{T}}(-J_s)\{-DX_{H_\lambda}(P_m)N_s + D_\nu N_s\} \\ &= -T_s. \end{aligned}$$

Thus $\langle T_{\mathrm{HL}}\rangle^{-1} = -\langle T_s\rangle^{-1}$, and their induced infinity norms agree. All entrywise analytic majorants and transpose norms used by the canonical constants are invariant under these fixed sign changes. This identifies the saved inverse bound with hypothesis H4; it does not identify the two torsion matrices themselves. No exact Lagrangianity of the approximate P_m is presumed.

Verification of the analytic theorem's hypotheses. For each fixed real $\lambda \in [0,1]$, the common real-type holomorphic state domain and the uniform field bounds in (S21) provide H1. Canonical geometric structures are constant, so their nonzero zeroth-order bounds are one and their positive-order derivative bounds are zero. Since $n = d = 2$, no additional first integrals are required; one may choose the auxiliary conserved function $c = 0$. Scalar Hamiltonian-gradient norms, if needed, are bounded by four times the corresponding maximum component field bound. The nonconstant mechanical inertia is retained inside H_λ and its field derivatives throughout.

The complete finite Fourier prefix, its exact conjugate reality, and its uniformly positive distance from the boundary of the common inner state domain give H2. Whole-strip complex Gram invertibility gives the full-rank condition H3. Its bound, and the separate bounds for L and L^{T}, receive the strict auxiliary margins σ_B, σ_K and σ_{K^t}. The actually certified continuous signed mean, transported for the same complete prefix and every real $\lambda \in [0,1]$, gives H4 under the sign mapping above. A finite parameter scan or a desired inverse target cannot replace this premise. The fixed exact period-one frequency has its all-integer bounded-continued-fraction lower bound; together with the uniform period-one cohomological estimate it gives H5 with $\tau = 1$ and the certified γ, c_R.

The complete ordinary parameter-series defect, including every computed coefficient and all infinite parameter and angle tails, supplies the theorem's error norm. For each real λ, the same uniform constants and all eight strict Case III tests therefore give the theorem's invariant map and its common C^0 correction bound. The corrected harmonic phase test separately proves that the real map is an embedding. Its derivative allowance uses the Cauchy loss on the final strip, including the factor $2\|S^{-1}\|\Delta/\rho_\infty$ in the derivative norm. Applying the theorem separately to each real parameter proves precisely the quantified per-parameter conclusion of (S25). It does not add analytic parameter dependence, a positive-measure set of frequencies, or a prescribed common energy level.

Identification of the reference endpoint. For the specified scalar graph, let $s = d^{\mathsf{T}}q$ and

$$V_4(q) = \tfrac{1}{2}q^{\mathsf{T}}K_bq + \tfrac{1}{2}s^2 - \tfrac{1}{24}s^4, \qquad U(q) = 1 - \cos s - \tfrac{1}{2}s^2 + \tfrac{1}{24}s^4.$$

The family is an exact identity of real-analytic functions,

$$H_\lambda = T_{G_8} + V_4 + (1-\lambda)N + \lambda U.$$

Consequently

$$H_1 = \tfrac{1}{2}p^{\mathsf{T}}G_8(q)^{-1}p + \tfrac{1}{2}q^{\mathsf{T}}K_bq + 1 - \cos(d^{\mathsf{T}}q).$$

Both N_q and N_p cancel at this endpoint; N_p is retained for every intermediate parameter and throughout the continuation certificates. The endpoint theorem concerns this prescribed finite-Gauss conservative reference. Correspondence with a

continuum rod, uncertain measured plant, dissipative trajectory or sampled controller requires the respective additional certificates.

This component is a conditional theorem application. Numerical acceptance requires a completed same-prefix whole-strip residual, the corresponding scalar assembly, and their independent final review; none is inferred from the present convention check.

S7 Numerical application to the full conservative family

The following application instantiates Theorem S11 and the canonical conventions above for the same selected 736-coordinate GC output and the same 129-dimensional finite observation certificate. It uses no additional fit or target coefficient as a training label. All stored network buffers and Fourier coefficients are interpreted as exact constants of their declared graph.

Actual prefix and uniform geometry. The base embedding K_0 is the 8064-mode rational Fourier polynomial used for the learned-system theorem. Each of the ten response polynomials U_j is computed from the complete preceding prefix, and the subsequent certificates use

$$P_{10}(\varphi,\lambda) = K_0(\varphi) + \sum_{j=1}^{10} \lambda^j U_j(\varphi).$$

The retained angle half-width is $\rho = 1/40$ and the outer width is $\widehat{\rho} = 1/10$. Uniform geometry, analytic Gram invertibility and the varying continuous signed mean are verified for every real $\lambda \in [0,1]$. The inverse mean bound is

$$\sup_{0\le\lambda\le1} \|\langle T_\lambda\rangle^{-1}\|_\infty < 25539.514.$$

It is obtained from the signed mean at the complete degree-three prefix and a certified transport through the remaining actual responses. It is not the inverse mean at K_0 or a finite parameter scan. The observation-derived first-, second- and third-order energy-error bounds enter the uniform field constants in (S21), with all momentum derivatives retained.

Complete residual series. Write the ordinary parameter series of the exact composed defect as

$$\mathcal{R}_{10}(\varphi,\lambda) = X_{H_\lambda}(P_{10}) - D_\nu P_{10} = \sum_{j\ge0} r_j(\varphi)\lambda^j.$$

Vector analytic norms here take the maximum of the component suprema on the whole retained angle strip. Table S1 gives upward-rounded bounds on $\|r_j\|_{1/40}$. The zeroth coefficient comes from the complete learned-field strip certificate, and the first from a complete 448×448 Fourier calculation. For each coefficient $2 \le j \le 14$, a complete 384×384 Fourier enclosure retains every sample ball and representative, including the Nyquist modes. The exact half-shift reconstruction, angular weights and all infinite angle aliases are included. An independent reverse-axis transform verifies all complete Fourier coefficient bounds. Separate raw-model replays at two actual grid points and one fresh nongrid point check the composed coefficient evaluator against the original neural and mechanical graph.

The finite degree of P_{10} does not truncate the nonlinear composed residual at degree ten. In particular its eleventh coefficient is retained. A direct product-domain certificate proves holomorphicity for $\varphi \in \mathbb{T}^2_{1/10}$ and $|\lambda| \le 512$, together with the residual majorant $M < 0.937873$. This certificate evaluates the full prescribed inertia and raw constrained neural graph; it does not assume containment in the older local state tube. Cauchy's estimate therefore gives

$$\sum_{j>14} \|r_j\|_{1/40} \le \frac{M}{512^{14}(512-1)} < 2.158 \times 10^{-41}. \tag{S29}$$

Consequently, uniformly for real $\lambda \in [0,1]$,

$$\|\mathcal{R}_{10}(\cdot,\lambda)\|_{1/40} \le \sum_{j=0}^{14} \|r_j\|_{1/40} + \frac{M}{512^{14}(512-1)} < 3.848 \times 10^{-41}. \tag{S30}$$

Table S1. Whole-strip ordinary parameter-coefficient bounds for the same complete P_{10}. Display rounding is upward; the sum is formed from the exact rational endpoints before rounding.

Order j	Upper bound on $\|r_j\|_{1/40}$
0	1.19552×10^{-42}
1	7.09445×10^{-43}
2	1.36625×10^{-43}
3	2.66122×10^{-46}
4	5.18925×10^{-49}
5	1.01274×10^{-51}
6	1.97800×10^{-54}
7	3.86361×10^{-57}
8	7.54577×10^{-60}
9	1.47356×10^{-62}
10	2.87754×10^{-65}
11	1.48568×10^{-41}
12	5.33047×10^{-45}
13	2.36871×10^{-48}
14	1.18583×10^{-51}

The full learning-error, geometry, domain and actual-mean inputs give a sufficient complete-defect budget greater than 9.303×10^{-40}.

Strict inequalities and embedding. The eight Case III inequalities are evaluated anew with the exact complete defect from (S30), rather than a pointwise residual or a desired target. Their upward-rounded values are in Table S2; each must be strictly less than one. The largest value is below 0.04136.

Table S2. The eight actual uniform strict tests for P_{10}.

Test	Upper bound
Quadratic contraction	4.13578×10^{-2}
Iteration allowance	1.02610×10^{-38}
Approximate symplectic inverse	7.45219×10^{-32}
DP_{10} preservation	1.07339×10^{-17}
DP_{10}^{T} preservation	1.07352×10^{-17}
Gram-inverse preservation	1.58011×10^{-16}
Mean-inverse preservation	2.18268×10^{-7}
Domain-distance preservation	2.15032×10^{-19}

The resulting uniform C^0 correction is $\Delta < 3.949 \times 10^{-22}$ on the final strip of width $1/400$. The remaining state-domain distance exceeds 0.001836. Let S_h be the fixed real invertible harmonic normalization used to construct K_0, and define

$$\kappa_a(\varphi) = a\big(\cos(2\pi\varphi_1), \cos(2\pi\varphi_2), -\sin(2\pi\varphi_1), -\sin(2\pi\varphi_2)\big)^{\mathsf{T}},$$
$$K_{\mathrm{harm}} = S_h \kappa_a, \quad a = \frac{5764607523034235}{144115188075855872} \simeq 0.04.$$

Put $s_h = \|S_h^{-1}\|_\infty$ and $\rho_\infty = 1/400$. The actual certified upper bounds obey

$$\eta_0 \geq \sup_{\lambda\in[0,1]} \|S_h^{-1} P_{10} - \kappa_a\|_{\rho_\infty}, \qquad \eta_1 \geq \sup_{\lambda\in[0,1]} \|D_\varphi(S_h^{-1} P_{10} - \kappa_a)\|_{\rho_\infty}.$$

These include the base harmonic remainder and all ten responses; they are upper bounds, rather than assumed exact norms. Define

$$b_0 = \eta_0 + s_h\Delta, \qquad b_1 = \eta_1 + \frac{2 s_h \Delta}{\rho_\infty}.$$

The added derivative allowance follows by Cauchy's estimate at the real torus, summing the two angle derivatives of the corrected map. The certified quantities satisfy $a > 2b_0$ and

$$\frac{2b_1 + (88/7)b_0}{6(a - 2b_0)} < 0.005686 < \frac{1}{2}.$$

Thus the corrected real maps are embeddings, as well as invariant maps.

Application and endpoint. The fixed exact Diophantine frequency, canonical structures, common analytic domain, full-rank geometry, actual varying mean and the strict inequalities now satisfy all hypotheses of Theorem S11. Applying it separately to every real $\lambda \in [0, 1]$ proves the invariant tori and common displacement bound in article Theorem 4. At $\lambda = 1$ both N_q and N_p cancel exactly, leaving the prescribed conservative finite-Gauss reference. This application uses the realized deterministic observation certificate; the independent-anchor statistical experiment is not required for this conclusion and has not been executed. The sampled dissipative plant is covered by its separate physical-energy proof.

S8 Complex-gradient descent with a strict descent margin

Theorem 3 in the published version of Zhang et al. (2022) motivates complex-gradient optimization, but a stationary-point conclusion requires a strict descent margin. The following convention-explicit statement also applies to a smooth CHNN regression objective.

Proposition S13. Complex descent. *Let $\mathcal{L} : \mathbb{C}^P \simeq \mathbb{R}^{2P} \to \mathbb{R}$ be bounded below and have an L-Lipschitz real gradient on every segment traversed by the iterates. For $w = a + ib$, define $G(w) = \nabla_a \mathcal{L} + i\nabla_b \mathcal{L} = 2\partial\mathcal{L}/\partial\bar{w}$. Choose a constant $\rho \in \mathbb{C}$ with $\kappa = \operatorname{Re}\rho - (L/2)|\rho|^2 > 0$, and update $w_{t+1} = w_t - \rho G(w_t)$. Then*

$$\mathcal{L}(w_{t+1}) \le \mathcal{L}(w_t) - \kappa\|G(w_t)\|_2^2, \qquad \lim_{t\to\infty} \|G(w_t)\|_2 = 0,$$

and

$$\min_{0\le t<T} \|G(w_t)\|_2^2 \le \frac{\mathcal{L}(w_0) - \inf \mathcal{L}}{\kappa T}.$$

Proof. Under realification, multiplication by $\rho = u + iv$ is $uI + vJ_c$, with $J_c^\top = -J_c$ and $J_c^\top J_c = I$. Hence the update has inner product $-u\|G\|^2$ with the real gradient and squared norm $|\rho|^2\|G\|^2$. The real descent lemma gives the first inequality. Summing it and using the lower bound for $\mathcal{L}$ gives $\kappa \sum_{t\ge 0} \|G(w_t)\|^2 \le \mathcal{L}(w_0) - \inf \mathcal{L}$. Square summability proves convergence of the gradient norm and the minimum bound. The result asserts convergence to stationarity in gradient norm, not convergence to a global minimum or to a particular parameter vector.

For a real step this condition is $0 < \rho < 2/L$ under the stated convention. The case $\rho = 0$, allowed by a non-strict printed step-size interval, cannot imply gradient convergence: for $\mathcal{L}(w) = |w|^2$ and $w_0 \neq 0$, every iterate equals w_0 and $G(w_t) = 2w_0 \neq 0$. Nonsmooth complex activations require separate treatment. The present literal complex $\tanh$ networks are smooth as real maps wherever their poles are excluded. On a bounded parameter region with a uniform pole margin, the smooth regression objective admits finite local Lipschitz bounds. Applying the proposition to a training sequence would still require a verified common iterate region and the strict step-size margin above.

This realification argument is an optimization component motivated by Zhang et al. (2022); it does not transfer that paper's exponential-loss learning conclusions to the present mechanical regression. The actual GC conservative fit uses 400 full-batch Adam updates on its 736 real parameter coordinates, with the previously fitted PSD matrix head fixed. Training-only checkpoint selection returns update 385. The independent matrix-head fit uses damped Gauss–Newton steps with Armijo search. Its fitting auxiliary quantities belong to that head-identification problem, not to the GC conservative objective. Neither algorithm is the constant-complex-step iteration of this proposition. The finite Adam record and the head's accepted decreases therefore do not establish the proposition's gradient-norm convergence conclusion. The identification-error and a-posteriori KAM results are independent of such convergence: they do not assume that the accepted frozen parameters are stationary or globally optimal.

S9 A conditional learning-to-analytic-KAM implication

We relate the energy generalization estimate of Chen et al. (2022) to the analytic invariance residual required by Haro and Luque (2019, Theorem 3.1). The argument combines an interior gradient estimate with interpolation on a complex torus strip. This conditional statistical route requires hypotheses on the sampling distribution and the entire hypothesis class. A deterministic a-posteriori certificate for an individual learned Hamiltonian has a distinct scope: it establishes the existence of an invariant torus of that Hamiltonian without a population-risk or generalization statement.

For the specified four-coordinate model the target is $H_\star = T_{G_8} + V_4 + U(q)$, and the learned energy is $H_\vartheta = T_{G_8} + V_4 + N_\vartheta(q,p)$. Their shared kinetic graph and canonical normalization are retained; in particular, no derivative $N_{\vartheta,p}$ is set to zero.

S9.1 The conditional statistical input and its energy gauge

Work in one fixed canonical coordinate and time normalization, with phase dimension $m = 2n$ and $n \geq 2$. Let $D \subset \mathbb{R}^m$ be a bounded, connected Lipschitz domain. The conservative target is $H_\star$, and $\mathcal{H} = \{H_\vartheta : \vartheta \in \Theta\}$ is a hypothesis class fixed before the data are drawn. All energies are real valued on D, even when the network parameters are complex. Fix a finite exponent $p > m$ and a confidence parameter $0 < \alpha < 1$. Assume $H_\star, H_\vartheta \in W^{1,p}(D)$, using their continuous representatives, and define the pointwise conservative loss

$$\ell_\vartheta(x) = \|\nabla H_\vartheta(x) - \nabla H_\star(x)\|_p^p. \tag{S31}$$

Assume that $X_1, \ldots, X_N$ are independent and identically distributed with law P supported on D, whose density relative to Lebesgue measure satisfies $f_P \geq m_P > 0$. Assume $0 \leq \ell_\vartheta \leq L_{\max}$ uniformly on the entire class and domain. Let $\overline{\mathfrak{R}}_N$ be a valid upper bound for the empirical Rademacher complexity of this loss class, with the normalization in Lemma 2 of Chen et al. (2022). Define

$$\begin{aligned} \widehat{L}_{p,N}(\vartheta) &= \frac{1}{N}\sum_{i=1}^{N} \ell_\vartheta(X_i), \\ A_N(\vartheta,\alpha) &= \widehat{L}_{p,N}(\vartheta) + 2\overline{\mathfrak{R}}_N + 3L_{\max}\sqrt{\frac{2\log(4/\alpha)}{N}}. \end{aligned} \tag{S32}$$

Under these assumptions the cited generalization bound supplies an event $\mathcal{G}_\alpha$, with $\mathbb{P}(\mathcal{G}_\alpha) \geq 1 - \alpha$, on which $\mathbb{E}_P \ell_\vartheta \leq A_N(\vartheta,\alpha)$ simultaneously for every $\vartheta \in \Theta$. This uniformity is essential when the same data select the fitted parameters.

Define the gauge-fixed error

$$e_\vartheta = H_\vartheta - H_\star - c_\vartheta, \qquad c_\vartheta = \frac{1}{|D|}\int_D (H_\vartheta - H_\star)\, dx. \tag{S33}$$

Let C_P satisfy $\|e\|_{L^p(D)} \leq C_P \|\nabla e\|_{L^p(D)}$ for mean-zero functions, and let C_S be a valid constant in $W^{1,p}(D) \hookrightarrow C^0(\overline{D})$, using $\|e\|_{W^{1,p}}^p = \|e\|_{L^p}^p + \|\nabla e\|_{L^p}^p$.

Proposition S14. Conditional energy bound. *On $\mathcal{G}_\alpha$, simultaneously for every member of the class,*

$$\begin{aligned} \|e_\vartheta\|_{C^0(\overline{D})} &\leq \varepsilon_\vartheta, \\ \varepsilon_\vartheta &:= C\left(\frac{A_N(\vartheta,\alpha)}{m_P}\right)^{1/p}, \\ C &= C_S(1 + C_P^p)^{1/p}. \end{aligned} \tag{S34}$$

Proof. The density lower bound and the uniform risk event give

$$\int_D \|\nabla e_\vartheta\|_p^p \, dx \leq m_P^{-1}\mathbb{E}_P \ell_\vartheta \leq m_P^{-1} A_N(\vartheta,\alpha).$$

Poincare's inequality applies because (S33) has zero mean. Consequently $\|e_\vartheta\|_{W^{1,p}} \leq (1 + C_P^p)^{1/p}(A_N/m_P)^{1/p}$. The Sobolev embedding, valid because $p > m$, proves the assertion.

The exponent $1/p$ in (S34) follows from the pth-power definition of the loss (S31); a linear dependence on that loss would require different definitions. For the four-dimensional phase space considered here, this first-order embedding requires $p > 4$. Here p is the integrability exponent in $W^{1,p}$, not a derivative order. Using weak-increment observations or a combined Hamiltonian–damping residual to bound $\widehat{L}_{p,N}$ additionally requires conservative-gradient label-error estimates, including errors in dissipation and input subtraction. The trajectory sampling scheme, with consecutive and overlapping windows, does not meet the iid state-sampling hypothesis of Proposition S14.

S9.2 A real gradient bound with explicit domain loss

Proposition S15. Interior gradient estimate from energy values. *Let* $e \in C^2(D)$ *satisfy* $\sup_D |e| \le \varepsilon$ *and* $\max_j \sup_D |\partial_{jj} e| \le B$, *with* $B \ge 0$. *Let* $S \subset D$ *and* $r > 0$ *satisfy* $x + [-r, r]\mathbf{e}_j \subset D$ *for every* $x \in S$ *and coordinate unit vector* $\mathbf{e}_j$. *Then*

$$\sup_{x \in S} \|\nabla e(x)\|_\infty \le b_r(\varepsilon, B) := \inf_{0 < h \le r} \left(\frac{\varepsilon}{h} + \frac{Bh}{2} \right). \tag{S35}$$

For $B > 0$ *this is*

$$b_r(\varepsilon, B) = \begin{cases} \sqrt{2B\varepsilon}, & 0 \le \varepsilon \le Br^2/2, \\ \varepsilon/r + Br/2, & \varepsilon > Br^2/2. \end{cases} \tag{S36}$$

For $B = 0$, *take* $b_r(\varepsilon, 0) = \varepsilon/r$.

Proof. For $0 < h \le r$, the fundamental theorem of calculus gives

$$\left| \partial_j e(x) - \frac{e(x + h\mathbf{e}_j) - e(x - h\mathbf{e}_j)}{2h} \right| \le \frac{1}{2h} \int_{-h}^{h} |\partial_j e(x + t\mathbf{e}_j) - \partial_j e(x)| \, dt \le \frac{Bh}{2}.$$

The difference quotient has magnitude at most ε/h. Taking the infimum proves (S35). For $B\varepsilon > 0$, the unconstrained minimizer is $h = \sqrt{2\varepsilon/B}$; otherwise use $h = r$ or take the appropriate zero limit. This proves (S36).

The estimate only needs bounds on the pure second derivatives of the *error*; a full Hessian operator bound is a stronger sufficient condition. A certified sum of target and network Hessian bounds also suffices. An additive energy constant has no effect on this step.

Let $J = \left(\begin{smallmatrix} 0 & I_n \\ -I_n & 0 \end{smallmatrix} \right)$, and fix a reference embedding $K_0 : \mathbb{T}^n \to D$ and frequency $\omega \in \mathbb{R}^n$. Suppose its real image has the coordinate-segment margin in Proposition S15, and that an independently justified target residual obeys

$$\|J\nabla H_\star(K_0) - DK_0\omega\|_0 \le \delta_\star. \tag{S37}$$

The subscript 0 denotes the supremum over real angles, with the maximum norm for vector components. Since $\|J\|_{\infty\to\infty} = 1$, the learned invariance residual satisfies

$$\begin{aligned} E_\vartheta &= J\nabla H_\vartheta(K_0) - DK_0\omega, \\ \|E_\vartheta\|_0 &\le \delta_\star + b_r(\varepsilon_\vartheta, B) =: \delta_\vartheta \quad \text{on } \mathcal{G}_\alpha. \end{aligned} \tag{S38}$$

In particular, a certificate for an invariant torus of H_ϑ cannot be substituted for the target residual in (S37). If K_0 is an exact torus of $H_\star$, then $\delta_\star = 0$; otherwise its defect must be included.

The actual finite-observation certificate also admits a deterministic application. Suppose Lemma S9 supplies the first-derivative bounds for the same frozen full-phase $H_\vartheta, H_\star$ on the real image $K_0(\mathbb{T}^n)$. Taking their physical-coordinate component maximum as in (S20) gives G_{det} and hence $\|E_\vartheta\|_0 \le \delta_\star + G_{\mathrm{det}}$ directly. The identical known kinetic metric in the two Hamiltonians is retained in this use of the deterministic theorem. Substitution of G_{det} for $b_r(\varepsilon_\vartheta, B)$ in the analytic step below gives a deterministic residual bound without a statistical assumption. Its applicability requires the actual configuration and momentum observation residuals, label fidelity, signed observation-point remainders, inverse correction

and target-domain remainder on the common domain. The GC conservative fit has no profiled potential nuisance. A small floating-point weak loss does not supply these hypotheses. No iid probability interpretation is attached to correlated windows.

S9.3 Analytic continuation of the residual with a certified outer bound

Write $\mathbb{T}^n_R = \{\theta \in \mathbb{C}^n/\mathbb{Z}^n : |\operatorname{Im}\theta|_\infty < R\}$. For a vector function, $\|F\|_R = \max_j \sup_{\mathbb{T}^n_R} |F_j|$. Assume that K_0 extends holomorphically to this strip and that $K_0(\mathbb{T}^n_R)$ lies in a pole-free complex domain of H_ϑ. Assume a *certified upper bound*

$$\|E_\vartheta\|_R \le M. \tag{S39}$$

For example, if verified constants satisfy $\sup_{K_0(\mathbb{T}^n_R)} \|J\nabla H_\vartheta\|_\infty \le C_X$ and $\|DK_0\|_R \le C_K$, then one may *choose* $M = C_X + C_K\|\omega\|_\infty$. Here the matrix norm is $\|A\|_R = \max_i \sum_j \sup_{\mathbb{T}^n_R} |A_{ij}|$, as in the analytic norm convention of Haro and Luque (2019). Point samples of a complex extension do not certify these suprema.

Lemma S16. Residual interpolation on a torus strip. *Let F be a bounded holomorphic periodic vector function on $\mathbb{T}^n_R$ with $\|F\|_0 \le \delta$ and $\|F\|_R \le M$. For $0 < \rho < R$, set $\bar\delta = \min\{\delta, M\}$. Then*

$$\|F\|_\rho \le \bar\delta^{\,1-\rho/R} M^{\rho/R}. \tag{S40}$$

If either bound is zero, the right-hand side is interpreted by continuity.

Proof. First suppose $M \ge \bar\delta > 0$ and that the extension is continuous on the closed strip. Given $x + iy$ with $t = |y|_\infty > 0$, put $v = y/t$. For each component define the one-variable function $g(w) = F_j(x + wv)$ on $0 \le \operatorname{Im} w \le R$. The real parts $x + (\operatorname{Re} w)v$ range over real angles, so the lower boundary has bound $\bar\delta$ and the upper boundary has bound M. The function is bounded throughout this strip. Set

$$\beta = R^{-1}\log(M/\bar\delta), \qquad G(w) = \bar\delta^{-1} g(w) e^{i\beta w}.$$

Both horizontal boundaries satisfy $|G| \le 1$. To justify the maximum principle on this unbounded domain, use $G_\eta(w) = G(w)\exp(-\eta w^2 - \eta R^2)$, with $\eta > 0$. Its modulus is at most one on the horizontal boundaries and tends to zero on the vertical boundaries of expanding rectangles. The rectangle maximum principle, followed by expansion and then $\eta \downarrow 0$, gives $|G(w)| \le 1$. In particular,

$$|F_j(x + iy)| = |g(it)| \le \bar\delta \exp(\beta t) = \bar\delta^{1-t/R} M^{t/R}.$$

Since $\bar\delta \le M$, this expression is nondecreasing in t. Taking $t \le \rho$ and the componentwise supremum proves the claim. For an open outer strip, apply the argument with $\rho < R' < R$ and let $R' \uparrow R$. Zero bounds follow by replacing the bounds with positive ones and taking a limit. This also covers real angles $t = 0$.

Combining the energy and gradient estimates with the strip lemma gives, on $\mathcal{G}_\alpha$, the analytic residual bound

$$\begin{aligned} &\|E_\vartheta\|_\rho \le \mathcal{E}_\vartheta, \\ &\quad \mathcal{E}_\vartheta := \bar\delta_\vartheta^{\,1-\rho/R} M^{\rho/R}, \\ &\quad \bar\delta_\vartheta := \min\{M, \delta_\vartheta\}, \end{aligned} \tag{S41}$$

where δ_ϑ is defined by (S38). There is no dimension factor for this maximum-component norm.

For the real energy readout of the complex network, analytic continuation means complexification of the real canonical variables, not application of a real-part operation at complex arguments. If $H(x) = \operatorname{Re} F(x)$ on real x and F has a holomorphic extension, its extension is $[F(z) + F^\#(z)]/2$, where $F^\#(z) = \overline{F(\bar z)}$ is a holomorphic function obtained by conjugating coefficients. For the actual network, the jth conjugate feature pair on real phase space becomes the independent linear expressions $(Q_j/s_j + iP_j/s_{2+j})/s_0$ and $(Q_j/s_j - iP_j/s_{2+j})/s_0$, $j = 1, 2$, after complexification. Here Q, P are physical canonical coordinates, $s_1, \dots, s_4$ are the four stored state scales, and s_0 is the stored binary64 normalization constant used by that network. Intermediate complex $\tanh$ layers must remain separated from their poles on the whole

declared complex domain. Bounds for one trained network do not establish corresponding uniform bounds for a parameter class.

S9.4 The precise a-posteriori KAM endpoint

We use the ordinary theorem of Haro and Luque (2019, Theorem 3.1, equations (3.9)–(3.11), arXiv v1), specialized to $d = n \geq 2$ and canonical geometry. This specialization has no additional first integrals; their constants are zero as in the cited Remark 3.2. In this paragraph $\mathcal{B}$ denotes the complex phase domain and is distinct from the scalar Hessian bound B. The following are the corresponding H1–H5 requirements, imposed independently of the empirical training loss:

- H1. The exact symplectic and metric structures and H_ϑ extend analytically to $\mathcal{B}$, with the specified analytic bounds. In canonical coordinates the geometric structures are constant, as in the cited Remark 3.3. The required Hamiltonian bounds concern $\|DH_\vartheta\|_{\mathcal{B}}$, $\|X_{H_\vartheta}\|_{\mathcal{B}}$, $\|DX_{H_\vartheta}\|_{\mathcal{B}}$, $\|D^2X_{H_\vartheta}\|_{\mathcal{B}}$ and $\|D(X_{H_\vartheta}^{\mathsf{T}})\|_{\mathcal{B}}$ in the cited tensor-norm convention. As emphasized in its Section 3.1, this last derivative of a row-vector function and the transpose of the Jacobian have different tensor-norm definitions and are not interchangeable. The optional first integral c of that theorem may be chosen constant, with zero derivative bound.
- H2. The embedding K_0 and DK_0 extend continuously to the closed strip of width ρ, are holomorphic inside it, and satisfy $\operatorname{dist}(K_0(\mathbb{T}^n_\rho), \partial\mathcal{B}) > 0$, $\|DK_0\|_\rho < \sigma_K$ and $\|(DK_0)^{\mathsf{T}}\|_\rho < \sigma_{K^{\mathsf{T}}}$.
- H3. The tangent frame $L = DK_0$ has maximal rank throughout the closed complex strip, and its metric Gram inverse $\mathsf{B} = (L^{\mathsf{T}}L)^{-1}$ satisfies $\|\mathsf{B}\|_\rho < \sigma_B$. The transpose is bilinear, not a Hermitian adjoint. The normal frame is defined by equations (3.2)–(3.5) of the cited theorem, with the same symplectic sign convention. For the present canonical application put $\Omega_{\mathrm{HL}} = -J$, $N_s = JL\mathsf{B}$ and $N_{\mathrm{HL}} = -N_s$. If $T_s = N_s^{\mathsf{T}}J(DX_{H_\vartheta}(K_0)N_s - D_\omega N_s)$, then $T_{\mathrm{HL}} = -T_s$ and $\|\langle T_{\mathrm{HL}}\rangle^{-1}\| = \|\langle T_s\rangle^{-1}\|$. This is the same sign identification as in the canonical conservative-family application; no exact Lagrangianity of the approximate embedding is presumed.
- H4. For the resulting torsion matrix T of equations (3.6)–(3.7), computed using X_{H_ϑ}, the real-angle average has an inverse satisfying $|\langle T\rangle^{-1}| < \sigma_T$. Small gradient error by itself does not establish this condition.
- H5. For fixed $\gamma > 0$ and $\tau \geq n - 1$, the frequency obeys $|k \cdot \omega| \geq \gamma|k|_1^{-\tau}$ for every $k \in \mathbb{Z}^n \setminus \{0\}$. A finite search for small divisors does not prove this all-integer assertion.

For $0 < \rho_\infty < \rho$, these data determine the explicit theorem constants C_1, C_2, C_3 of the cited Appendix B. Its smallness condition is exactly

$$\frac{C_1\|E_\vartheta\|_\rho}{\gamma^4\rho^{4\tau}} < 1. \tag{S42}$$

Accordingly, checking

$$\frac{C_1\mathcal{E}_\vartheta}{\gamma^4\rho^{4\tau}} < 1 \tag{S43}$$

and H1–H5 supplies, on $\mathcal{G}_\alpha$, an invariant analytic map K_∞ for the learned conservative Hamiltonian with the prescribed frequency, analytic on the strip of width ρ_∞. The cited theorem also gives

$$\|K_\infty - K_0\|_{\rho_\infty} \leq \frac{C_2\|E_\vartheta\|_\rho}{\gamma^2\rho^{2\tau}} \leq \frac{C_2\mathcal{E}_\vartheta}{\gamma^2\rho^{2\tau}}. \tag{S44}$$

To conclude that the corrected real map is an embedding, impose in addition a sound injectivity and immersion criterion for that corrected map. If the right-hand side of (S44) is Δ, its derivative on the real torus differs from DK_0 by at most $n\Delta/\rho_\infty$ in the induced infinity norm, by the coordinatewise Cauchy estimate. The real-embedding certificate must include this derivative allowance and the C^0 allowance Δ. A positive harmonic-amplitude margin and strict normalized phase test provide such a certificate for the four-coordinate learned instance. The corresponding criterion for a conservative continuation must be verified uniformly for its own complete embedding prefix and correction bound. Full rank of the approximate complex tangent frame alone is not an injectivity certificate for the corrected real map. Under this additional condition, the real restriction of K_∞ is an embedded invariant torus.

The zero-residual case is already invariant and admits $K_\infty = K_0$; the weak inequality also covers this case. The constants and geometric bounds must correspond to this Hamiltonian, domain and pair of strip widths. The conclusion concerns a

conservative torus, not attraction, dissipative control, all trajectories, or a positive-measure family without an additional family argument.

Proposition S17. Deterministic certification of an arbitrary training output. *Let* A *be any possibly randomized training and checkpoint-selection procedure, and let* $\widehat{\vartheta} = \mathsf{A}(\mathcal{Z}, \xi)$ *be its finite output parameters, including data-dependent preprocessing constants. For each realized dataset* $\mathcal{Z}$ *and random seed* ξ*, consider the exact exported Hamiltonian* $H_{\widehat{\vartheta}}$ *and any candidate* K_0*, frequency* ω *and verification domain. Suppose these objects satisfy H1–H5 and the corrected real-embedding criterion above in the same coordinate, time and norm conventions, and a valid deterministic bound* $\overline{E} \geq \|J\nabla H_{\widehat{\vartheta}}(K_0) - DK_0\omega\|_\rho$ *satisfies*

$$\frac{C_1\overline{E}}{\gamma^4\rho^{4\tau}} < 1.$$

Then $H_{\widehat{\vartheta}}$ *has the invariant torus and correction bound stated above, with* $\overline{E}$ *in place of* $\mathcal{E}_\vartheta$*. No sampling-independence, descent, stationarity or optimality condition on the training output is required.*

Proof. Fix an arbitrary realization $(\mathcal{Z}, \xi)$ and its exact output. The verified hypotheses and residual bound satisfy Haro and Luque (2019, Theorem 3.1), which gives the asserted torus and correction estimate. This implication holds for every realization; no property of the procedure producing the candidate is used.

This result concerns acceptance of a particular frozen function. It requires verified bounds for that function, rather than class-wide statistical constants, and does not guarantee that the training procedure produces an acceptable candidate. The finite-observation identity of Lemma S9 and its field increments (S20) likewise apply to arbitrary selected parameters under their stated observation and remainder hypotheses.

S9.5 An explicit risk threshold and the probability statement

For $B > 0$, $M > 0$, set

$$s = \rho/R, \qquad \kappa = \frac{\gamma^4\rho^{4\tau}}{C_1}, \qquad T_0 = \left(\frac{\kappa}{M^s}\right)^{1/(1-s)}. \tag{S45}$$

If $T_0 > \delta_\star$, define the positive admissible energy error

$$\varepsilon_{\mathrm{allow}} = \min\left\{\frac{Br^2}{2}, \frac{(T_0 - \delta_\star)^2}{2B}\right\}. \tag{S46}$$

A sufficient empirical-risk and complexity condition is

$$A_N(\vartheta, \alpha) < m_P\left(\frac{\varepsilon_{\mathrm{allow}}}{C}\right)^p. \tag{S47}$$

Indeed, (S34) then yields $\varepsilon_\vartheta < \varepsilon_{\mathrm{allow}} \leq Br^2/2$ and hence $\delta_\star + \sqrt{2B\varepsilon_\vartheta} < T_0$. Raising to $1 - s$ and multiplying by M^s proves (S43). The general criterion (S43) remains available when $B = 0$ or the optimal difference step is limited by r; these cases are covered directly by (S43).

Corollary S18. Uniform event and post-fit acceptance. *Suppose the statistical assumptions preceding Proposition S14 hold. Let* $\widehat{\vartheta}$ *be any data-dependent selection from the fixed class. Define acceptance to require valid deterministic bounds for all analytic, reference-residual and geometric assumptions above, including the corrected real-embedding certificate, together with* (S43) *evaluated using* (S34)*. Let* $\mathcal{A}$ *be this acceptance event and* $\mathcal{K}$ *the event that the asserted learned invariant torus exists. Then*

$$\mathbb{P}(\mathcal{A} \cap \mathcal{K}^c) \leq \alpha. \tag{S48}$$

Proof. On the uniform event $\mathcal{G}_\alpha$, the energy estimate applies to every parameter, including $\widehat{\vartheta}$. All subsequent implications are deterministic whenever the acceptance conditions hold. Thus erroneous acceptance is contained in $\mathcal{G}_\alpha^c$, whose probability is at most α.

Equation (S48) is not the claim $\mathbb{P}(\mathcal{K} \mid \mathcal{A}) \geq 1 - \alpha$. If the acceptance probability is positive, the immediate conditional bound is instead at most $\alpha/\mathbb{P}(\mathcal{A})$ for failure. An unconditional success probability of at least $1 - \alpha$ requires the remaining

conditions to hold for the selected model; a procedure that can reject that model gives only the joint-event guarantee above. Application of this statistical result requires uniform parameter-class bounds, a valid complexity estimate for the realified complex architecture, iid state samples with a positive density lower bound, and conservative pointwise label-error estimates. The correlated weak-window sampling scheme does not satisfy the iid hypothesis. Moreover, bounds for the fitted Hamiltonian are model-specific and do not provide the class-wide complexity or distributional constants required in (S32). The numerical trajectory-based results are therefore interpreted through their deterministic estimates, without a population-confidence claim for the fitted model. The conditional statistical learning-to-KAM route in this section has not been instantiated by the present experiments.

Remark S1. Necessity of quantitative regularity bounds. Real energy closeness alone does not control a vector field: on a bounded real interval, $e_k(q) = k^{-1}\sin(k^2 q)$ has sup norm tending to zero and derivatives of unbounded amplitude. Even a real Hessian bound does not supply (S39): the entire, 2π-periodic functions $e_k(q) = k^{-2}\sin(kq)$ have $\|e_k\|_0 \to 0$, $\|e_k'\|_0 \to 0$ and $\|e_k''\|_0 \le 1$, while $|e_k'(i\rho)| = k^{-1}\cosh(k\rho)$ diverges at any fixed $\rho > 0$. A pole-free extension together with a quantitative outer bound, not analyticity alone, is what makes Lemma S16 useful. These examples concern scalar functions and do not assert unboundedness within a separately constrained finite network class.

S10 Sampled physical-energy safety using a full-phase learned model

Let the physical reference and the frozen learned energy be

$$H_*(q,p) = \tfrac{1}{2}p^{\mathsf{T}}G(q)^{-1}p + V_*(q), \qquad \widehat{H}(q,p) = H_*(q,p) + e(q,p), \qquad G = SS^{\mathsf{T}} \succ 0.$$

The same canonical state and dimensionless time are used in every bound. The physical plant is

$$q' = v_* = H_{*,p}, \qquad p' = -H_{*,q} - R_* v_* + g(q)u, \qquad R_* = SM_*S^{\mathsf{T}}, \quad M_* \succeq 0, \tag{S49}$$

where $g(q) = d\cos(d^{\mathsf{T}}q)$. A separately fitted, configuration-only head supplies $\widehat{M}(q) \succeq 0$ and $\widehat{R} = S\widehat{M}S^{\mathsf{T}}$. Define

$$\widehat{v} = \widehat{H}_p, \quad e_p = \widehat{v} - v_*, \quad y_* = g^{\mathsf{T}}v_*, \quad \widehat{y} = g^{\mathsf{T}}\widehat{v}, \quad d_* = v_*^{\mathsf{T}}R_*v_*, \quad \widehat{d} = \widehat{v}^{\mathsf{T}}\widehat{R}\widehat{v}.$$

On a fixed real domain $\mathcal{D}$, assume sound same-model certificates

$$|e| \le \varepsilon_H, \qquad |e_{p,j}| \le \varepsilon_{p,j}, \qquad \|\widehat{M} - M_*\|_2 \le \delta_M. \tag{S50}$$

These are certificates for the two specified frozen functions; they do not assert joint retraining or identification of an unknown physical plant.

Transport of a learned sample to physical power. At a real state put

$$z = S^{\mathsf{T}}\widehat{v}, \quad \eta = S^{\mathsf{T}}e_p, \quad a \ge \|z\|_2, \quad b \ge \|\eta\|_2, \quad \bar{m} \ge \|\widehat{M}\|_2.$$

Valid choices include

$$a^2 \ge \widehat{v}^{\mathsf{T}}G\widehat{v}, \qquad b^2 \ge \varepsilon_p^{\mathsf{T}}|G|\varepsilon_p,$$

where $|G|$ is the entrywise absolute value. Then

$$|y_* - \widehat{y}| \le \sum_j |g_j|\varepsilon_{p,j} \le \sum_j |d_j|\varepsilon_{p,j}, \qquad d_* \ge \widehat{d} - 2\bar{m}ab - \delta_M(a+b)^2. \tag{S51}$$

Indeed, $S^{\mathsf{T}}v_* = z - \eta$ and

$$\begin{aligned} d_* &= (z-\eta)^{\mathsf{T}}\widehat{M}(z-\eta) + (z-\eta)^{\mathsf{T}}(M_* - \widehat{M})(z-\eta) \\ &\ge z^{\mathsf{T}}\widehat{M}z - 2\bar{m}ab - \delta_M(a+b)^2. \end{aligned}$$

The nonnegative term $\eta^{\mathsf{T}}\widehat{M}\eta$ was discarded. Thus no term involving e_p is removed by identifying the physical and learned velocities.

Let an observation box $C_k \subset \mathcal{D}$ contain the true sampled state. Suppose an interval computation encloses $\widehat{y}(C_k)$, provides a lower bound $\widehat{d}_k^{-}$ for $\widehat{d}$ throughout C_k, and upper bounds $\bar{a}_k, \bar{b}_k, \bar{m}_k$ for the quantities above. It may return rational values $\widetilde{y}_k, e_{y,k}, \underline{d}_k$ with

$$[\widehat{y}_k^{-}, \widehat{y}_k^{+}] + [-E_{p,k}, E_{p,k}] \subseteq [\widetilde{y}_k - e_{y,k}, \widetilde{y}_k + e_{y,k}], \qquad E_{p,k} \geq \sup_{C_k} \sum_j |g_j| \varepsilon_{p,j},$$

$$0 \leq \underline{d}_k \leq \max\{0, \widehat{d}_k^{-} - 2\bar{m}_k \bar{a}_k \bar{b}_k - \delta_M(\bar{a}_k + \bar{b}_k)^2\}. \tag{S52}$$

Consequently $|y_*(t_k) - \widetilde{y}_k| \leq e_{y,k}$ and $\underline{d}_k \leq d_*(t_k)$. Directed rounding of the error upward and of the credit downward preserves these inequalities. For a real symmetric head, its maximum absolute row sum is a valid choice for $\bar{m}_k$.

Guard and physical derivative hypotheses. Choose $0 < c < E_g$. Assume that the designated physical-energy guard component $\mathcal{G} = \{H_* \leq E_g\}$ is compact and strictly contained in the state and clearance domain on which all required reference bounds hold. In particular, it lies in $\mathcal{D}$, with enough margin for the declared observation boxes. For every constant actual command $|u| \leq u_{\max}$, assume

$$|y_*'| \leq L_y, \qquad |d_*'| \leq L_d \tag{S53}$$

up to the first guard exit. Such bounds concern only the unchanged physical reference. Previously certified bounds on a larger guard can be used after independently proving this guard's inclusion. No derivative bound for a former learned Hamiltonian or former command record is used.

Quantized sampled filter. Initially use a sound energy upper bound

$$H_*(x_0) \leq W_0 = \left\lceil \max\{0, \widehat{H}^{+}(x_0) + \varepsilon_H\} \right\rceil_{\Delta_W} < c,$$

where $\lceil \cdot \rceil_{\Delta_W}$ rounds upward on the ledger grid. At later samples a certified fresh bound of the same form may decrease a predicted ledger by taking their minimum. Fresh evaluation is optional: retaining the valid predicted ledger suffices. No fresh reset is assumed in the present numerical record construction.

Suppose $H_*(t_k) \leq W_k < c$, and choose an exact mathematical hold length $h_k > 0$ and $0 < \alpha < 1$. Admit a nonzero *actual* command only if $|u_k| \leq u_{\max}$ and

$$I_k(u_k) := h_k(u_k \widetilde{y}_k + |u_k| e_{y,k} - \underline{d}_k) + \tfrac{1}{2} h_k^2 (|u_k| L_y + L_d) \leq \alpha(c - W_k). \tag{S54}$$

The command may be binary64, interpreted as its exact represented rational number. It is chosen inside an exact feasible interval and the inequality is checked for that represented value. Rounding an otherwise admissible real command toward zero is not by itself a valid substitute. For a nonzero command define

$$W_{k+1} = \lceil \max\{0, W_k + I_k(u_k)\} \rceil_{\Delta_W} \quad \text{and require} \quad W_{k+1} < c. \tag{S55}$$

If this final check fails, or there is no representable nonzero feasible command, use the separately admissible command $u_k = 0$ and $W_{k+1} = W_k$. The numerical realization uses $\alpha = 1/2$, $u_{\max} = 1/10$ and $\Delta_W = 2^{-80}$.

Safety conclusion and proof. If every observation enclosure and each actual hold satisfy the stated hypotheses, then

$$H_*(t_k + s) \leq \max\{W_k, W_{k+1}\} < c \quad (0 \leq s \leq h_k), \qquad H_*(t_{k+1}) \leq W_{k+1}. \tag{S56}$$

For a nonzero hold, physical power cancellation gives $H_*' = u_k y_* - d_*$. Before a possible first guard exit,

$$H_*(t_k + s) \leq W_k + sB_k + \tfrac{1}{2} s^2 A_k, \quad B_k = u_k \widetilde{y}_k + |u_k| e_{y,k} - \underline{d}_k, \quad A_k = |u_k| L_y + L_d \geq 0.$$

Since $s^2 \leq h_k s$, this quadratic is bounded by the affine line joining W_k to $W_k + I_k(u_k)$. Clipping a negative endpoint to zero and rounding upward can only raise that line. Its two endpoints are strictly below c. For $u_k = 0$, use instead the exact

physical identity $H'_* = -d_* \le 0$; no derivative reserve or learned credit is required. The strict inequality $c < E_g$ rules out the first exit and induction establishes (S56). If the sample times have no finite accumulation and the physical vector field is locally Lipschitz on the containing compact guard, bounded commands and compactness give forward completeness. This last statement remains conditional on successful sound sample evaluations at every future sample; a finite record is only a finite-horizon instance.

If only an upper bound $0 < h_k^{\mathrm{act}} \le \bar{h}_k$ on the actual holding time is known, use $\bar{h}_k$ in the gate but carry

$$W_{k+1}^{\mathrm{carry}} = \max\{W_k, W_{k+1}^{(\bar{h}_k)}\}.$$

The same affine argument then bounds the whole hold and its possibly earlier endpoint. Carrying only the endpoint at $\bar{h}_k$ is unsound when predicted power is negative and the hold ends earlier. The present mathematical protocol instead specifies exactly $t_k = k/1000$; binary64 ODE time tags construct numerical centres and do not replace that exact clock in an independent true-IVP validation.

Relation to a learned-energy ledger. The absence of an e_q term in the sampled gate follows from using the *physical* identity $H'_* = uy_* - d_*$. It is not an assertion that $e_q = 0$. A direct $\widehat{H}$ ledger along the physical plant requires

$$\widehat{H}' = -\widehat{d} + u\widehat{y} + e_q^{\mathsf{T}}\widehat{v} + e_p^{\mathsf{T}}(\widehat{R}\widehat{v} - \widehat{H}_q) + \widehat{v}^{\mathsf{T}}(\widehat{R} - R_*)v_*.$$

Its zero-input derivative need not be nonpositive. For example, take $H_* = (q^2+p^2)/2$, $\widehat{H} = q^2/2 + p^2$, $R_* = \widehat{R} = 1$ and $g = 1$. At $q = 0$, $p = r \neq 0$, $u = 3r/2$, the physical derivative is $H'_* = r^2/2 > 0$, whereas $-\widehat{d} + u\widehat{y} = -r^2$. Thus substituting the learned velocity for the physical velocity without (S51) can certify an incorrect power inequality.

Fixed-model instance and scope. The current implementation combines the actual full-phase GC energy with the separately trained configuration-only complex PSD head. The GC enters the initial energy bound, the learned port, the learned dissipation and the nominal request; it is not replaced by the prior velocity $G^{-1}p$. The prospective guard uses $c = 3/2000$ and $E_g = 1001c/1000$, subject to its separate domain certificate, and the finite example specifies 200 exact holds of length $1/1000$. Its sinusoidal reference has amplitude 0.003 and frequency 0.5; no tracking-convergence conclusion is asserted. Safety permits arbitrary finite nominal requests after filtering, so the particular tracking gains are not hypotheses of this theorem.

The 200-hold example concerns a finite-dimensional numerical physical reference, not hardware execution or joint reidentification. Its state boxes must independently contain the true reference samples. Ordinary ODE solutions, agreement of two numerical tolerances, or monitoring a few interior points does not establish this premise. Actuator error, uncertain physical time, delay, external force and sensor error require their own budgets. The distributed-viscosity coefficient remains a declared numerical assumption rather than a measured calibration. The conservative KAM result for this GC does not supply these control hypotheses, and neither earlier controllers nor their executed command sequences are transferred to the new composition.

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